

Security

SAP offers different tools, **processes**, and measures for security check to protect the data. Sap security helps to ensure that users can only use the functionality of sap which is a part of their job.

Security is a balancing act **for protecting** the sap data and its applications from unauthorized use and access.

It refers to the implementation of measures to protect the system, data, and processes **from unauthorized access**, misuse, or any damage.

There are **Two** Types of security:

- **Internal security** = managed by **SAP authorization concept**, access is controlled at the application level.
- **External security** = managed by **underlying database system**, access is controlled at the database level.

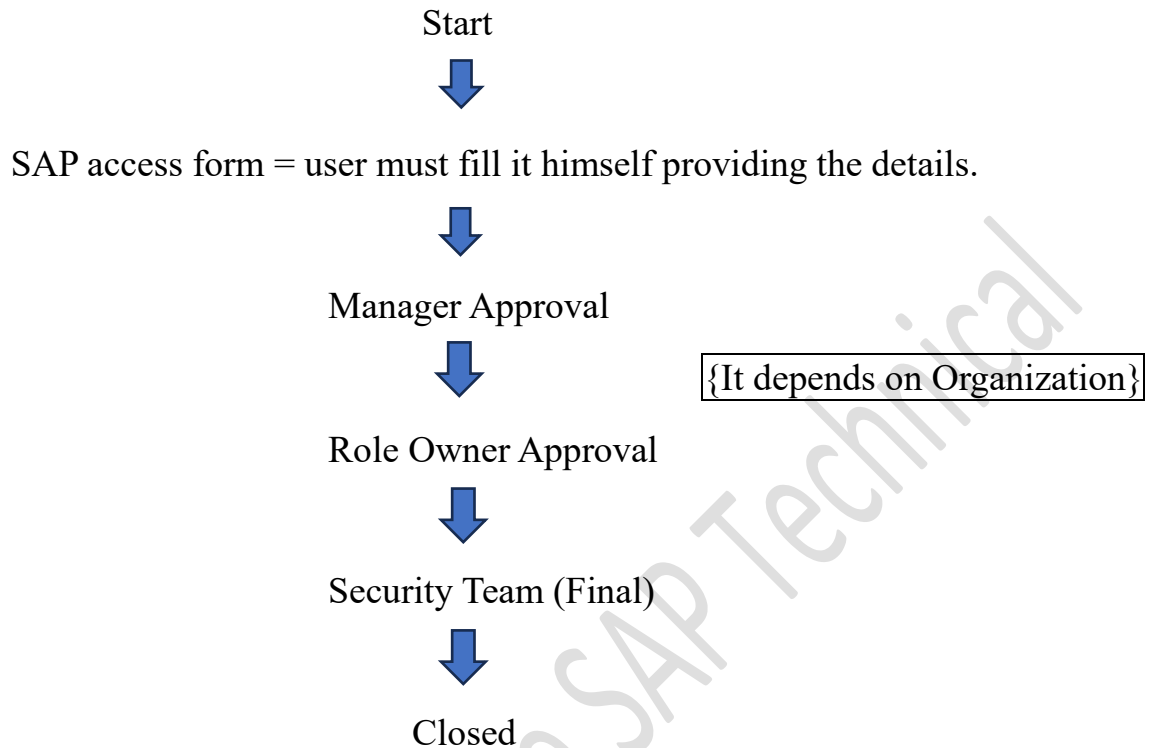
USER ADMINISTRATION

❖ It is all related to **User Maintenance**.

- 1) User creation
- 2) User modification
- 3) User deletion
- 4) User Lock and Unlock
- 5) User copy
- 6) User Password reset.

SU01 = > T code **for User Administration**

Process involved in the User Administration



SU01 Tabs

*Create *Modify *Display *Delete *Copy *Lock/Unlock *Change Password

< SAP User Maintenance: Initial Screen

✓ [varun] [icon] [icon] [icon] [icon] [icon] [icon] ... Cancel

User [varun] [icon]

Alias []

✚ When you click on **Create** total **11 tabs** will be visible.

✓

▼

📄

🔗

🔄

Cancel

User

VARUN

Changed By

SAPUSER

05.04.2024

09:11:12

Status

Saved

Documentation

Address

Logon Data

SNC

Defaults

Parameters

Roles

Profiles

Groups

Personalization

Lic. Data

➤ Documentation = enter the **description** and document for the user.

Description

Person Responsible

Documentation for User:

➤ Under address data only **Last Name** is Mandatory.

Person

Title	Mr. ▼
Last name	VARUN
First name	
Academic Title	

➤ Under Logon data **User type** and **initial password** is Mandatory.

Documentation	Address	Logon Data	SNC	Defaults	Parameters	Roles	Profiles	Groups	Personalization	Lic. Data
Alias										
User Type	Service									
Security Policy										
Password										
New Password Rules (Case-Sensitive)										
New Password										
Repeat Password										
Password Status	Production Password									

- SNC (Secure network Communication) = **Skip the password** if active. If not asking the password.

Documentation	Address	Logon Data	SNC	Defaults	Parameters	Roles	Profiles	Groups	Personalization	Lic. Data
SNC Status										
 SNC is not active on this application server  Unsecured logon is permitted for specific users										

- **Default** means common to every user like date format, decimal notation, time format, printer details, time zone.

Start menu	
Logon Language	
Decimal Notation	1.234.567,89
Date Format	DD.MM.YYYY
Time Format (12/24h)	24 Hour Format (Example: 12:05:10)
Spool Control	
Output Device	LP01
☐ Print Now ☐ Delete After Output	
Personal Time Zone	
Time Zone	CET

- Parameters = Only required to be entered when requested by the **User**.

Parameters



SET/GET Parameter ID	Parameter value	Short Description
☐		
☐		
☐		

- Roles = Assign authorization through **Roles**.

Role Assignments



Status	Role	Ty...	Start Date	End Date	Short Role Description	In
☐						
☐						

- Profiles are interlinked with **roles**, and they get assigned automatically once the roles are assigned.
- Maximum number of profiles that can be assigned to one user = **312**.

Assigned Authorization Profiles

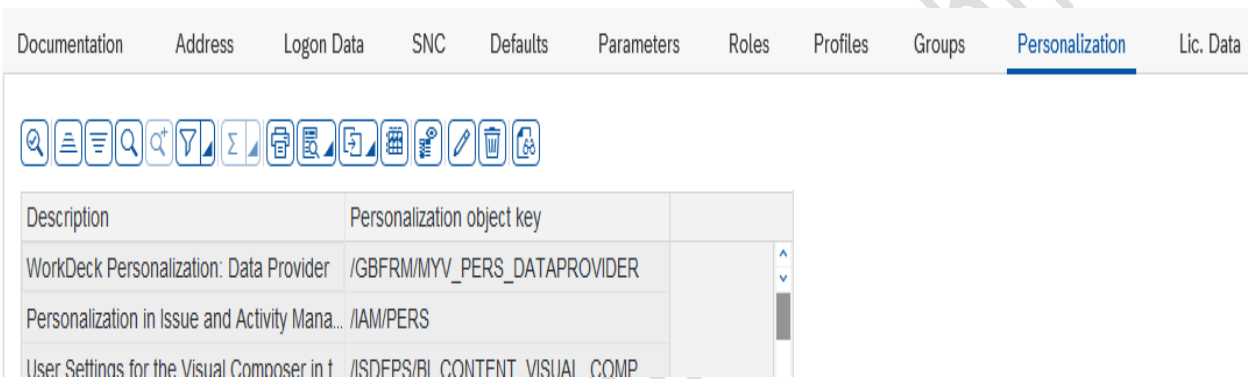


Profile	Type	Text
☐		
☐		

- Groups = assign **user** to extra group if he belongs to them.



➤ Personalization = do not enter any details here.



➤ License data = this is related to licensing of users. Admin, developers, etc.



Contractual User Type ID

❖ Whenever you are copying the user, **Address** and **password** will not copy.

Change docs = To see changes done to the user in a time interval

Path: SU01 ➡ information ➡ Change documents for Users

USER TYPES IN SU01

Under logon data, **initial password** (mandatory), it also consists of **user type**, user group, validity period, etc.

There are five user types.

- | | |
|------------------|--------------|
| 1) Dialog | 4) Service |
| 2) System | 5) Reference |
| 3) Communication | |

✚ **Dialog user type** = default user type, interactive user, password parameters are applied, Gui login is allowed.

Ex => All employees of the organization.

✚ **Service user type** = used for multiple dialog logons, password parameters are not applied, Gui login is allowed.

Ex => FFID in Grc.

✚ **System user type** = used for internal communication, system related activities, password parameters are not applied, Gui login is **not** allowed.

Ex => Background jobs are scheduled by using these System ids.

- ✚ **Communication user type** = used for external communication, between the system or across the landscape, password parameters are not applied, Gui login is **not** allowed.

Ex => RFC (Remote function call)

- ✚ **Reference user type** = used for providing extra access to dialog ids, when it accesses limit (312 profiles) is reached, no password required to login, cannot login through Gui.

Ex => Create reference user and assign **two** roles. and create **one** service user assign the **one** role and also assign the reference user to this user in the role tab, then one service user has access to **three** roles.

DIFFERENT TYPES OF USER LOCKS / SAP UFLAG VALUE

- It is the numbering value to determine the status of a user in the SAP system.
- The status here means, lock/unlock mode which can be refer to the following details:

0 = Not Locked

32 = Global Lock (Cua)

64 = Administrator Lock

128 = Incorrect logon Locks

128 + 64 = 192 => Password Lock + Locked by Administrator

128 + 32 = 160 => Password Lock + Locked by Cua

Standard /default users created after SAP Installation:

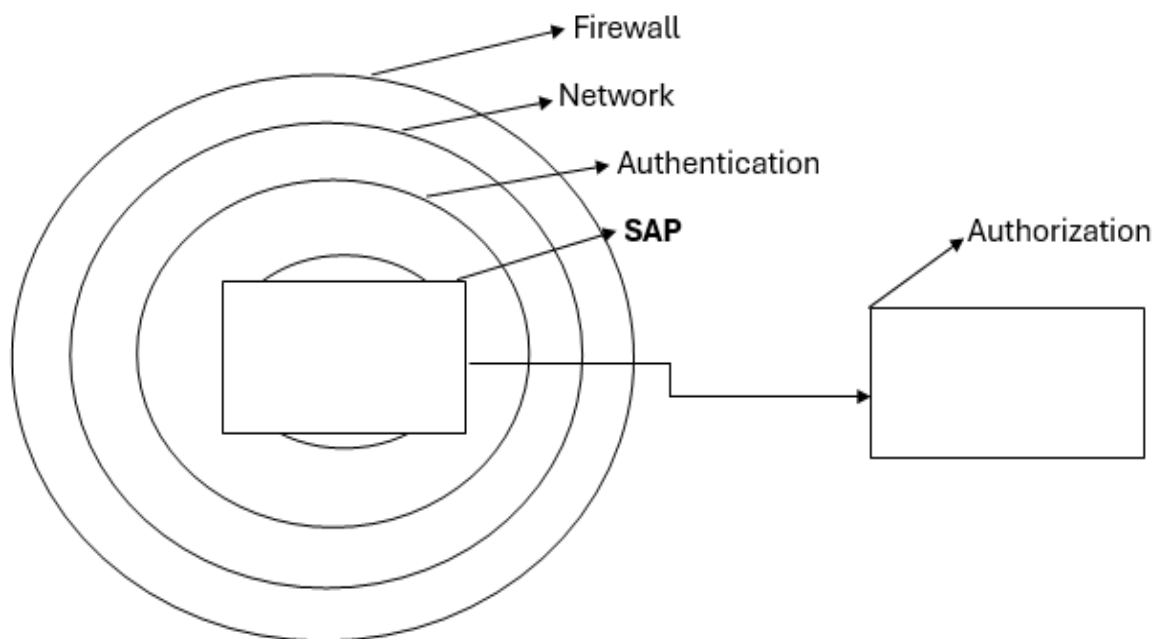
User id	Password	Client
Sap*	PASS/06071992	ALL
ddic	19920607	000, 001

Some facts

- ❖ User id maximum length = 12 characters. It differs from organization to organization based on the combination of first and last name.
- ❖ Maximum no of profiles assigned to one user = 312.
- ❖ Mandatory fields for creating the user are last name and initial password.
- ❖ All the tables related to user starts with USR*

Ex => USR02 ↘ User logon data.
 ↘ User illegal password.

AUTHORIZATION CONCEPTS



In the above diagram, there will be many outer layers, in order to login to SAP System you need to have break this firewall and the network. As a sap security consultant, we don't deal with firewall and the network.

We deal with **Authentication** to login to sap system by providing the login credentials (**user id, password**). After entering SAP system whatever did inside is Called **Authorization**.

- Authorization exists in inside system; Authentication exists in outside system.
- If someone know the user id and password another user also login to sap system, so the strong authentication check will be **biometric**.
- Authorization deals with permissions and privileges, authentication is something it checks your identity. So, **security** deals with both.
- Authentication is common to every user; authorization will be varied for every system.
- Authorization is identified by auth field and auth objects.
- Authorization objects is a group of **10** auth fields maximum.

Auth objects

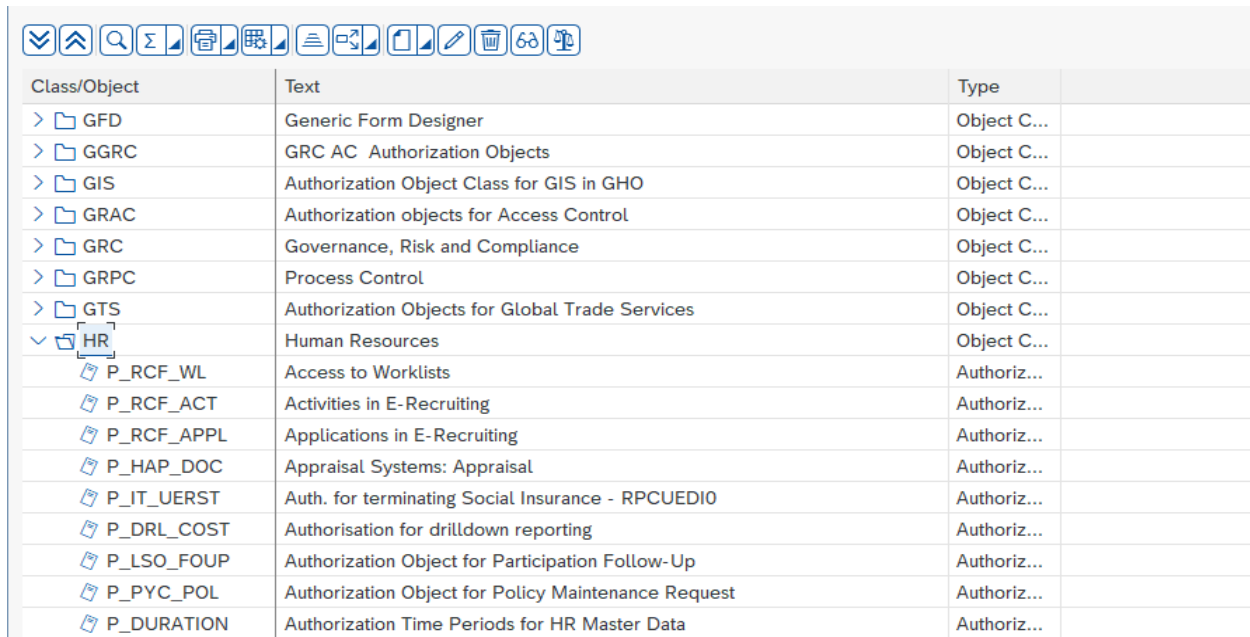
Auth field 1	Value
AF2	V2
AF3	V3
-	-
-	-
-	-
-	-
-	-
AF10	V10

- SU21** = T code for auth objects.
- Object class reveals the area of authorization objects. (Each folder belongs to 1 area in that files are objects.

Class/Object	Text	Type
> []	Objects with Invalid Object Class	Object C...
> [] /F15	/F15	Object C...
✓ [] AAAA	Obsolete Authorization Objects	Object C...
[] S_EDITOR	ABAP: Editor Checks	Authoriz...
[] P_CA	Academic Calendar (Obsolete - Do Not Use)	Authoriz...
[] RSDDLTP	Analytical Index Maintenance (Obsolete, Use S_RS_AI)	Authoriz...
[] C_SCP_MODR	Assigning Resource Allocation Hierarchies to Planning Models	Authoriz...
[] S_TSKH_ADM	Authorization Check for SM04, SM50	Authoriz...
[] P_GB_UCAS	Authorization Object UCAS (obsolete, do not use)	Authoriz...
[] E_PRICE	Authorization Object: Price	Authoriz...
[] M_SC_EKGRP	Authorization for Admin SC	Authoriz...

For example, = HR member come and ask, what is the authorization object for appraisal, payroll, and recruitment, first we need to find out the **authorization object**.

Path = SU21 => scroll down => HR (Human resources)



The screenshot shows the SAP authorization object list. At the top, there is a toolbar with various icons for navigation and actions. Below the toolbar is a table with three columns: 'Class/Object', 'Text', and 'Type'. The 'HR' object is selected and expanded, showing a list of sub-objects. The sub-objects are listed with their respective text and types.

Class/Object	Text	Type
> GFD	Generic Form Designer	Object C...
> GGRC	GRC AC Authorization Objects	Object C...
> GIS	Authorization Object Class for GIS in GH0	Object C...
> GRAC	Authorization objects for Access Control	Object C...
> GRC	Governance, Risk and Compliance	Object C...
> GRPC	Process Control	Object C...
> GTS	Authorization Objects for Global Trade Services	Object C...
✓ HR	Human Resources	Object C...
P_RCF_WL	Access to Worklists	Authoriz...
P_RCF_ACT	Activities in E-Recruiting	Authoriz...
P_RCF_APPL	Applications in E-Recruiting	Authoriz...
P_HAP_DOC	Appraisal Systems: Appraisal	Authoriz...
P_IT_UERST	Auth. for terminating Social Insurance - RPCUEDIO	Authoriz...
P_DRL_COST	Authorisation for drilldown reporting	Authoriz...
P_LSO_FOUP	Authorization Object for Participation Follow-Up	Authoriz...
P_PYC_POL	Authorization Object for Policy Maintenance Request	Authoriz...
P_DURATION	Authorization Time Periods for HR Master Data	Authoriz...

Below activities are performed under SU01

To Create User => S_USER_GRP is auth object.

Assign roles to user => S_USER_AGR is auth object.

Assign profiles to user => S_USER_PRO is auth object.

Role 1 => SU01 => S_USER_GRP + S_USER_AGR

(user cannot assign profile to another user)

Role 2 => SU01 => S_USER_AGR + S_USER_PRO

(user cannot create another user)

Role 3 => SU01 + No auth objects

(user cannot perform any activity under SU01) means inside SU01 or any tabs whatever we do that is controlled by **auth objects**.

Note = necessary authorization object required for particular T code, have been linked with **SU24**.

SU24 => T code to find the authorization object related to any t code in SAP.

Path => SU24 => enter t code = **SU01** {any T code} => we get authorization object related to a particular T code. (filter on proposal by yes)

Transacti...	Short Description	Authorization Object	Object Descripti...	TSTCA	Check Ind.	Proposal Status	Package	Component ID
SU01	User Maintenance	☐ S_TCODE	Transaction Co...		Check	No	SUSR	BC-SEC-USR...
		☐ S_TRANSLAT	Translation envi...		Check	No	STRO	BC-DOC-TTL
		☐ S_TRANSPRT	Transport Orga...		Check	No	SCTS_...	BC-CTS-ORG
		☐ S_USER_AGR	Authorizations: ...		Check	Yes	S_PRO...	BC-SEC-AUT...
		☐ S_USER_AUT	User Master Mai...		Check	Yes	SUSR	BC-SEC-USR...
		☐ S_USER_GRP	User Master Mai...		Check	Yes	SUSR	BC-SEC-USR...
		☐ S_USER_PRO	User Master Mai...		Check	Yes	SUSR	BC-SEC-USR...
		☐ S_USER_SAS	User Master Mai...		Check	Yes	SUSR	BC-SEC-USR...
		☐ S_USER_SYS	User Master Mai...		Check	Yes	SUSR	BC-SEC-USR...
		☐ S_USER_TCD	Authorizations: ...		Check	No	S_PRO...	BC-SEC-AUT...
		☐ S_USER_VAL	Authorizations: ...		Check	No	S_PRO...	BC-SEC-AUT...
		☐ S_WFAR_OBJ	ArchiveLink: Aut...		Check	No	SWI	BC-BMT-WFM

For example, = users come to you and asking that I want access to sales order creation, printer, payroll. => VA01, SP01, PA30

- ❖ First you need to ask for that T codes, once you aware of that, now pull the auth objects related to these t codes than combine.
- ❖ After, create one single role and assign these above 3 t codes + auth objects related to these 3 t codes (group of t codes and its objects related to these t codes) = **Role**.

Note => we cannot assign T codes and authorization objects directly to the users. Hence, we are placing them under a role and then the role is assigned to user.

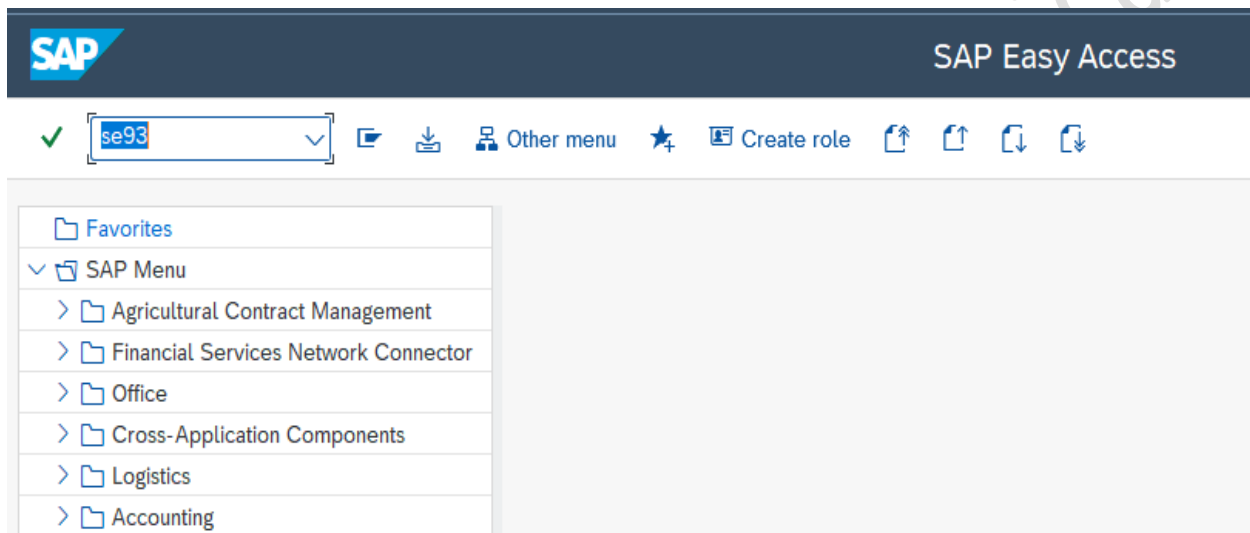
T Code => It is an easily memorable code to navigate to the desired program without the long names.

- ✓ T code acts as a shortcut to a program.

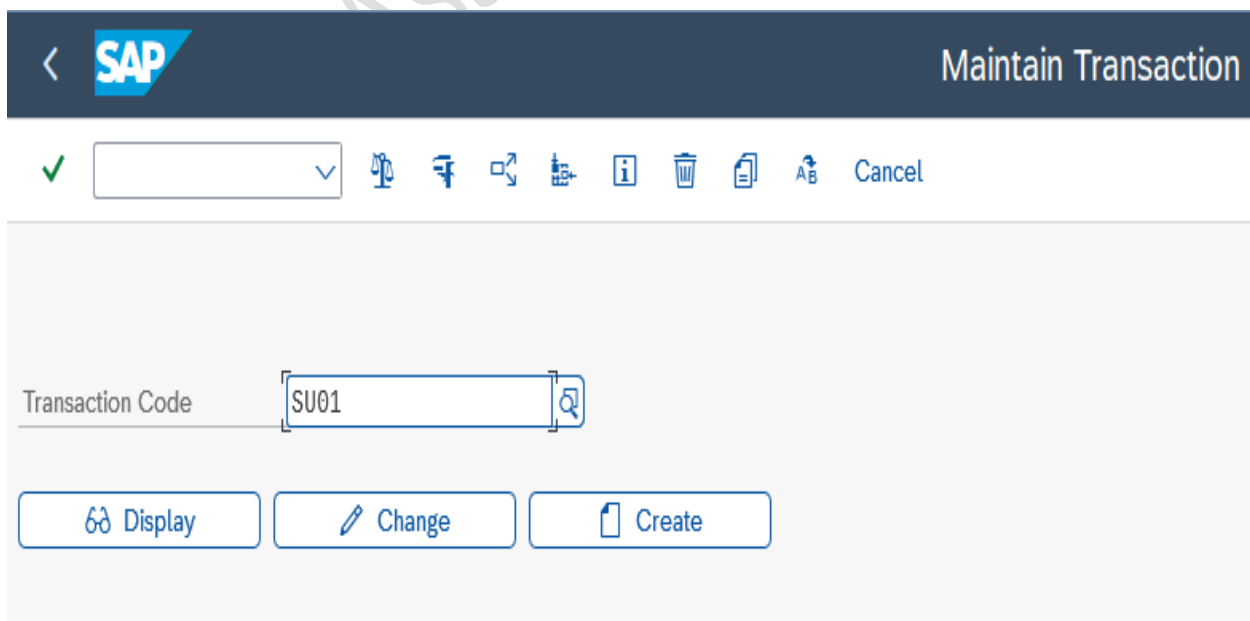
- ✓ Whenever you are executing any t code the system as hitting in the backend to show you the screen => **T code**.
- ✓ **SE93** => maintenance of all t codes
- ✓ **SE93** is a t code where you can find, **create**, maintain, display, delete and modify, etc.

How to find what is the program related to the particular T code means

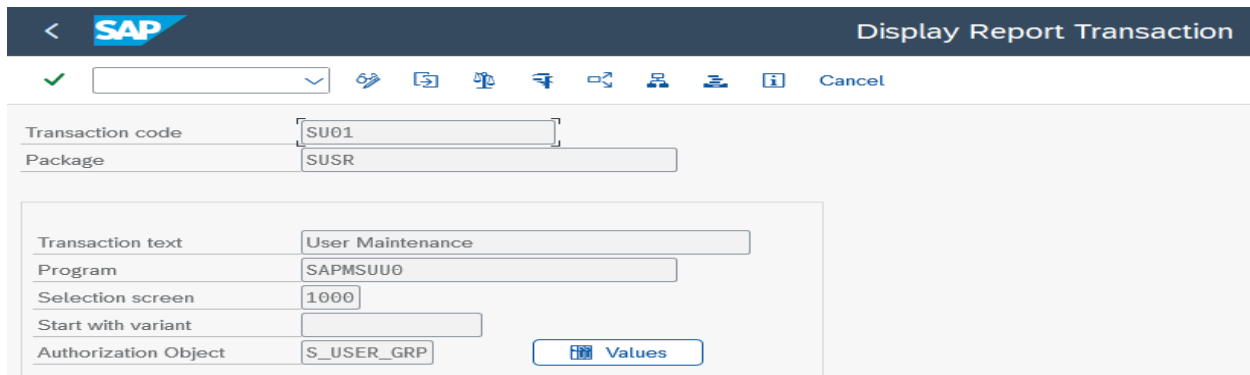
- Execute a T code called **SE93** in a command field and click on enter.



- To find a program or auth object related to SU01 (any t codes), type the T code in a transaction code field.



- Then click on display it will show the program and auth object related to a particular t code.

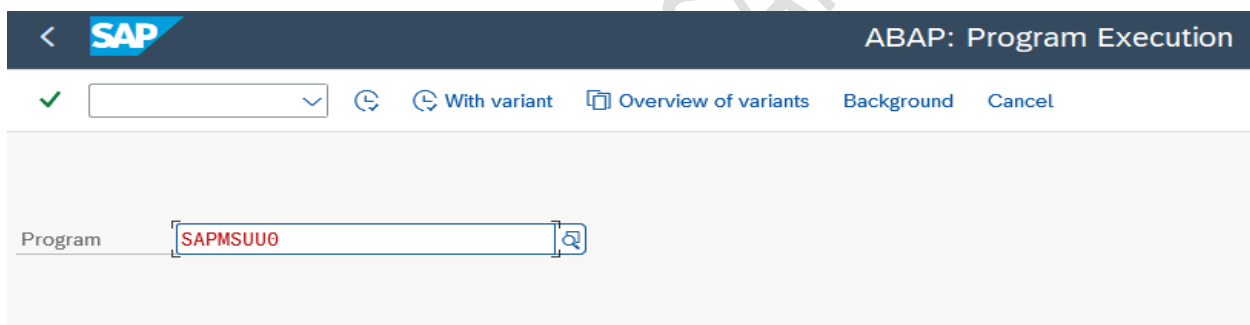


The screenshot shows the 'Display Report Transaction' screen in SAP. The transaction code 'SU01' is entered in the 'Transaction code' field, and the package 'SUSR' is entered in the 'Package' field. Below these fields, a table displays the following information:

Transaction text	User Maintenance
Program	SAPMSUU0
Selection screen	1000
Start with variant	
Authorization Object	S_USER_GRP

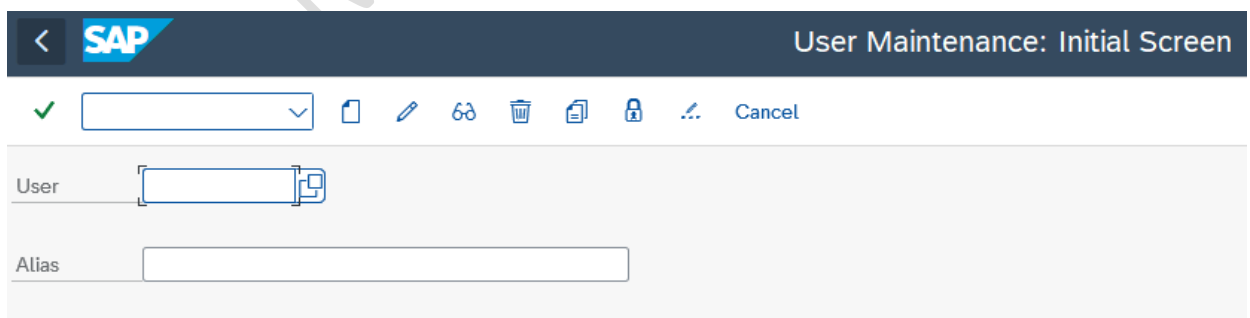
A 'Values' button is located next to the Authorization Object field.

- Here the program name is **SAPMSUU0** and the auth object is **S_USER_GRP**.
- ❖ To execute this program, go to a Command field and enter a t code called **SA38** and enter a program name **SAPMSUU0**.



The screenshot shows the 'ABAP: Program Execution' screen in SAP. The program name 'SAPMSUU0' is entered in the 'Program' field. The screen includes a toolbar with options like 'With variant', 'Overview of variants', 'Background', and 'Cancel'.

- ❖ Click on above execute, it will be getting the screen of **SU01**. (User maintenance).



The screenshot shows the 'User Maintenance: Initial Screen' in SAP. The 'User' field is empty, and the 'Alias' field is also empty. The screen includes a toolbar with options like 'Cancel'.

Note => hence it is very difficult to remember the program instead of this they put a shortcut code called T code.

SU01 => program name = **SAPMSUU0**, SCC5 => program name = **SAPMSCC1**

PROFILE and ROLE Concept

PFCG => profile generator

Profiles are of **two** types.

- 1) Standard profiles.
- 2) Generated thru rules.

- **Standard** means which comes along with the installation. It can be assigned to **user directly**.



	Profile name	Text
☐	A_ALL	FI-AA Asset Accounting: Full authorization
☐	A_ANZ	FI-AA Asset Accounting: Complete display authorization
☐	A_BEW_ALL	Asset Chart of Depreciation - All
☐	A_IMA_ALL	Appropriation Requests: All Authorizations
☐	A_IMPR_00	Investment Programs: Master Data, Budget and Planning
☐	A_IMPR_01	Investment Programs: Budget and Planning
☐	A_IMPR_ALL	Investment Programs: All Authorizations
☐	A_IMPR_DIS	Capital investment programs: All display authorizations
☐	A_IMPR_SYS	Investment Programs: System Administrator
☐	A_PROFIL_00	FI-AA Asset Accounting: Profile for system administrator
☐	A_PROFIL_01	FI-AA Asset Accounting: Profile for asset accountant
☐	A_PROFIL_02	FI-AA Asset Accounting: Profile for asset accounting clerk

- SAP recommends never assign the standard profiles to **users**. As they give extra access. Hence, we assign **roles** to users which in turn assigns the generated profiles to the users.
- **Generated** profiles cannot be assigned to **user directly**.
- Generated profiles are always starts with **T**.

Assigned Authorization Profiles

Profile	Type	Text
☐ SAP_NEW		New authorization checks
☒ T-W8750044		

Ty... Message Text

T-W8750044 is a generated profile. Assignment is therefore impossible.

- When you assign the **roles** to the users, the profiles will get automatically assigned.

ROLE

- It is a group of **T codes** and its related **authorizations**.

For example, => Create a necessary role for each and every team in the project.

Following teams are there in the project, finance, HR, sales, basis, security, abap.

- **HR team** performs below activities.

* Personal administration => PA20, PA30, PA40

* Organizational management => PO10, PO13, PPOM

Roles are categorized as two Types:

- **Technical roles** => support roles (create the software's)
- **Functional roles** => end users (use the software's and prepare the bills)

Role matrix

- Information **gathered from** all the technical teams, regarding the roles and the types of roles and which access they needed.
- With in the **business roles**, in Sales department there can be various levels of employees like clerk, supervisors, manager, senior manager.

For example, => Sales team - sales support role, salesclerk role, sales supervisor role, sales manager role.

Role creation / Modification process

- Change **request form** has to be filled by the respective technical team.
- That has to be approved by the CAB (**change advisory board**).
- Once approved, the requests come's to **security** and we create the role.

Profile concept.

PFCG => Profile generator / Role maintenance

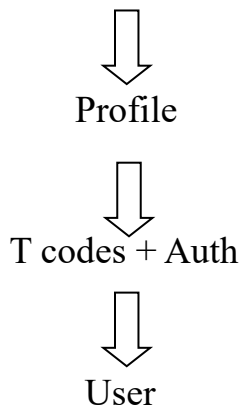
The screenshot displays the SAP Role Maintenance screen. At the top, the SAP logo and 'Role Maintenance' title are visible. Below this is a toolbar with various icons and the text 'Transactions' and 'Cancel'. The main content area features a 'Role' field containing 'X:BS:ABAP_DEVELOPMENT:TATA', a 'Short Description' field, and buttons for 'Single Role' and 'Comp. Role'. At the bottom, there are icons for 'Views' and a 'Show Documentation' button.

Old versions of SAP are SAP 3.0, 4.60, 4.66, 4.7, ECC5, ECC6.....

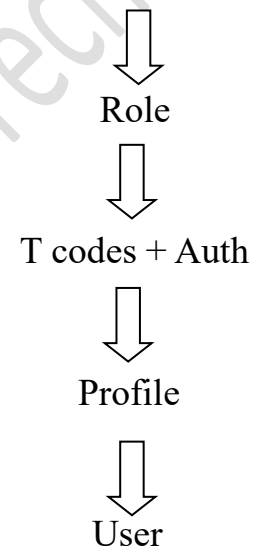
- **In 4.7 version** => Major changes occurred in security area, **role** concept has been introduced from **4.7** onwards, prior to that **profile** concept was there

IN 4.7 changes that happened

SU02 => to create profile



After that, PFCG



Role naming conventions

Types of roles => Single (X), Composite (Y), Derived (Z) /child, Parent (P) /Master

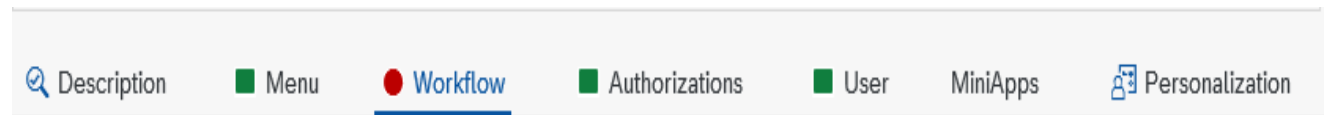
Role Length => Maximum **30** characters

Role name should contain the following: -

- ✓ Role type
- ✓ Business process => basis, Fico, HR
- ✓ Sub process => HR (PA, PD, Payroll)
- ✓ Extra information => Client

Example, => **X:HR:PERSONAL_ADMIN:TATA**

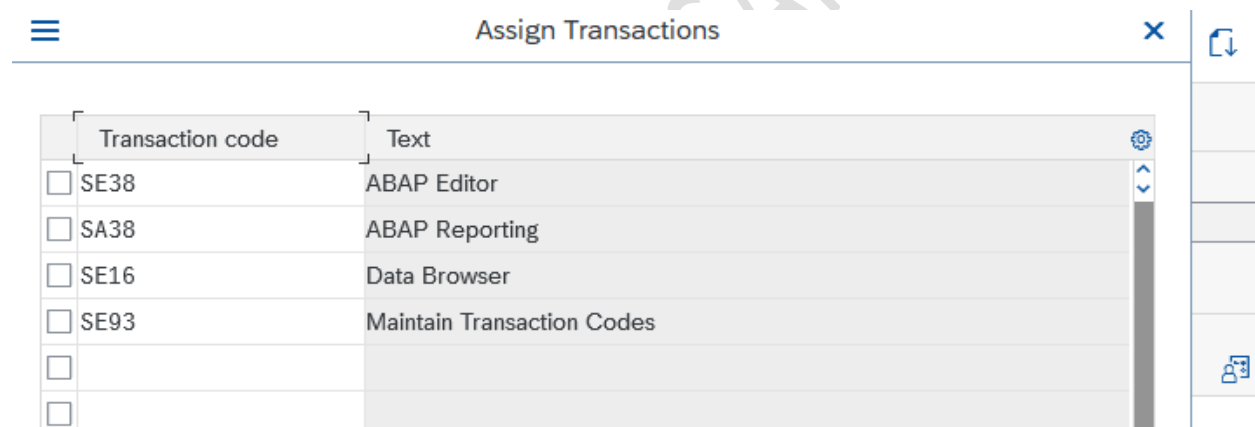
PFCG Tabs



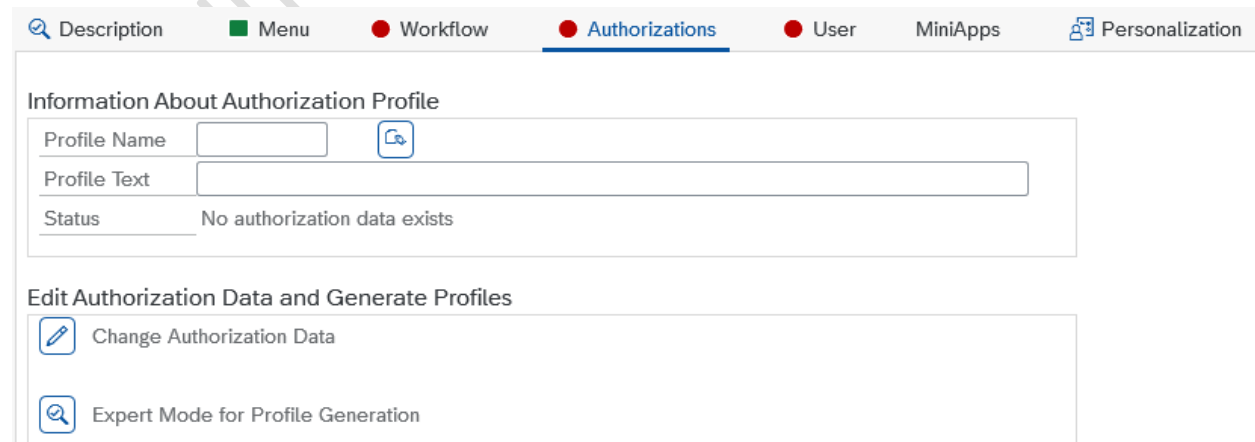
⇒ **Description** => just fill the long text with the information like why, when and who is creating the role.



⇒ **Menu** => Fill the list of T codes in the Transactions option.



⇒ **Authorization Tab** => go to change auth data and generate profile.



How to see what does each **field** and each **color** represents in authorization tab means:

Path => Utilities => Click on **Legend** or More => **Legend**

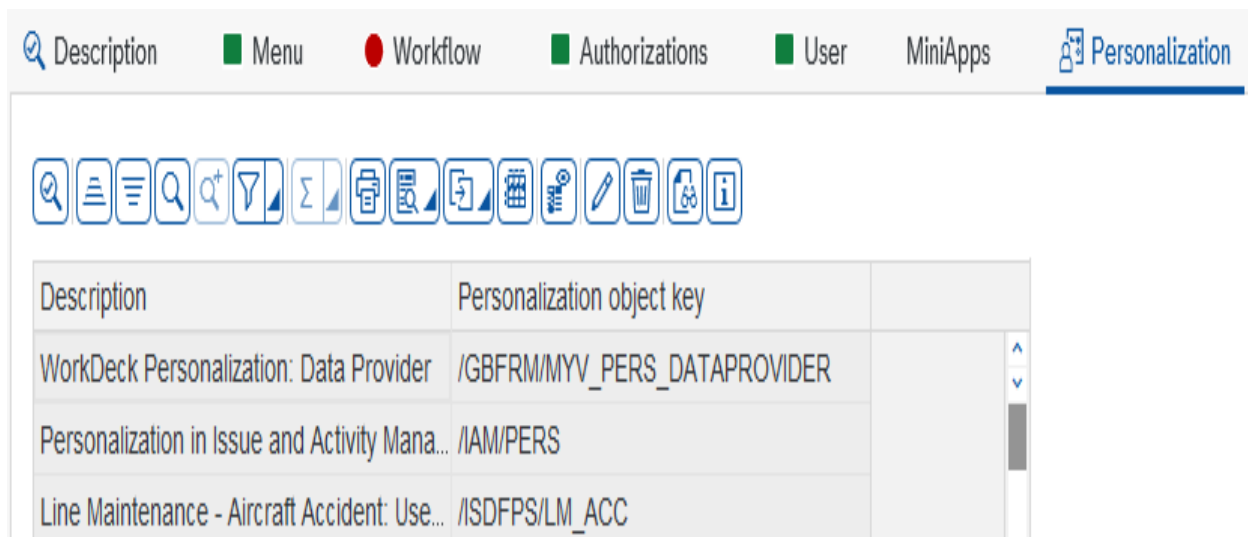
The screenshot shows the SAP 'Change Role: Authorizations' interface. On the left, a tree view displays authorization objects under the path 'X:BS:ABAP_DEVELOPMENT:TATA'. The objects are color-coded: orange for 'Cross-application Authorization Objects', green for 'Transaction Code Check at Transaction Start', and yellow for 'Basis: Administration'. The legend on the right lists various icons and their meanings, such as 'Object class', 'Authorization Objs', 'Authorizations', 'Authorization fids', 'Fields changed', 'Field conts', 'Technical names', 'Authorization field content maintained', 'Some authorization field content not maintained', 'Unmaintained organizational levels', 'Maintain Field Content', 'View field contents', 'Delete field contents, or delete an inactive authorization, or delete further authorizations for an object', 'Copy authorization', 'Deactivate an active authorization, or authorizations for an object', 'Transactions for an authorization object', and 'Reactivate an inactive authorization, or authorizations for an object'.

⇒ **User Tab** => indicates the list of users and assigned to the role, click on user comparison, once done you will see the user comparison tab become **green**.

The screenshot shows the 'User Assignments' table in SAP. The table has columns for 'User ID', 'User Name', 'From', 'to', and 'I...'. The first row shows 'TESTID' for 'varun' with dates '10.06.2024' and '31.12.9999'. Below the table, there are buttons for 'Selection', 'Organizational Mg...', 'User Comparison', and 'Info'. The 'User Comparison' button is highlighted in green.

User ID	User Name	From	to	I...
TESTID	varun	10.06.2024	31.12.9999	

⇒ **Personalization tab** => Do not touch it.



TESTING METHODS

Positive Testing

- Checking whether can executes the T codes under the role.

Negative Testing

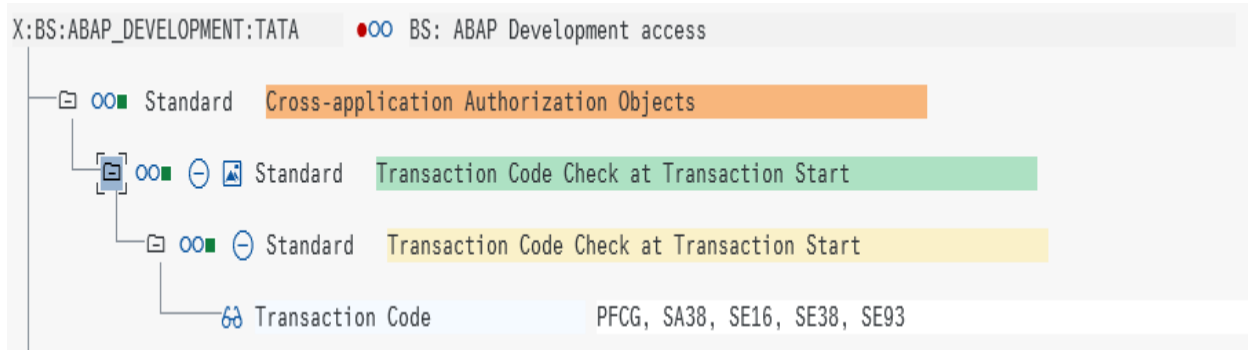
- Ensuring the user is not executing other T codes, except which are not under the role.

Authorization Tab

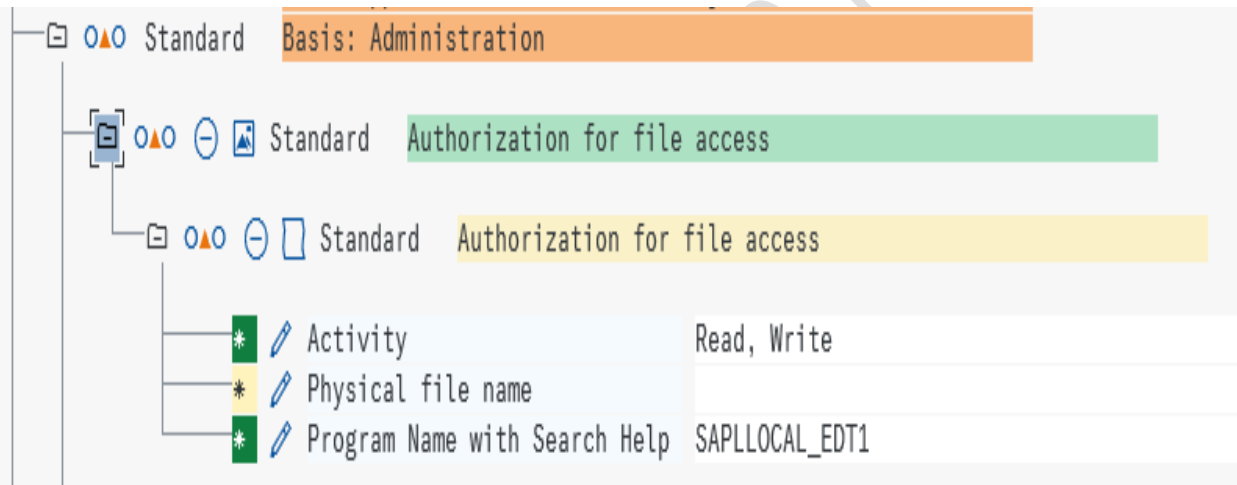
Traffic Lights

○○■	Authorization field content maintained
○▲○	Some authorization field content not maintained
●○○	Unmaintained organizational levels

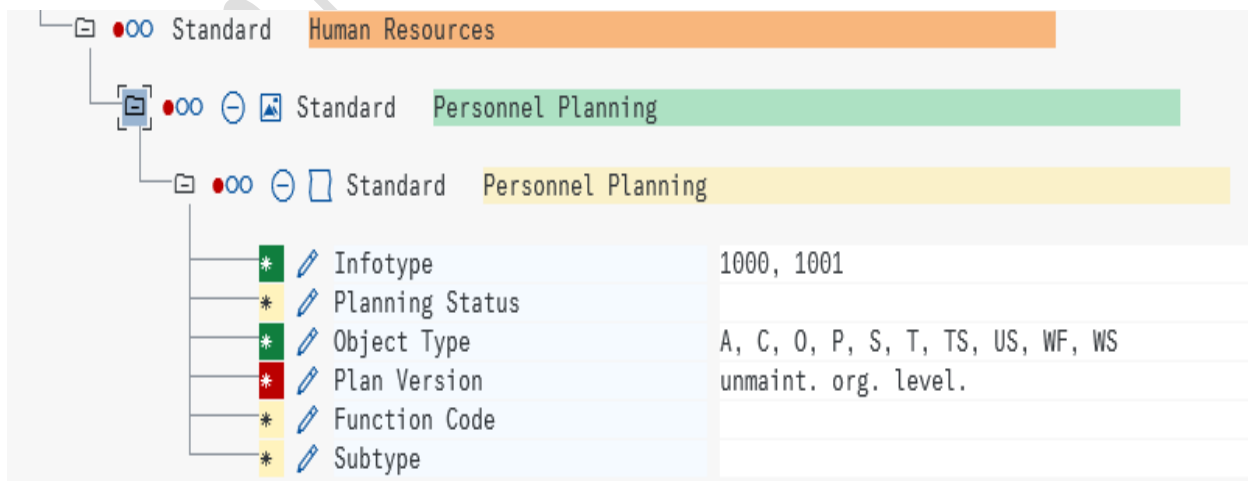
- ✓ **Green** => All authorization fields are **filled** or maintained with **values**.



- ✓ **Yellow** => At least **one** authorization field is not maintained or **partially** unmaintained.



- ✓ **Red** => unmaintained **organization levels**.



Organizational field

- * Authorization objects contain **authorization fields**.
- * Authorization objects is a group of **ten** authorization fields maximum.

Display authorization object

Object	S_GUI
Text	Authorization for GUI activities
Class	BC_A Basis: Administration
Author	SAP

Authorization fields

Authorization Field	Short Description...
ACTVT	Activity

It is of **Two** types: -

❖ **Normal field** => The field which does not represents **organization**.

Example, => Activity, role name

* Activity	33, 34	ACTVT
* Physical file name		FILENAME
* Program Name with Search Help	SAPLOCAL_EDT1	PROGRAM

❖ **Organizational field** => The field which represents **organization**.

Example, => Plant, cost center, billing

Define the values for the organizational levels of the role.

Field Vals of OrgLevels

Org. Level	'From'	'To'	More...	Change
☐ Company Code				
☐ Controlling Area				
☐ Plan Version				
☐ Division				

Org field Value

- Meaning varies from organization to organization means **specific to organization**.

* BMW => plant (manufacturing hub): 100(UK plant), Walmart => plant: 100(USA branch).

- Whereas normal field value is **universal**, Example => Activity: **03 (Display)**.

Object: S_TABU_DIS Table Maintenance (using sta...)

Field name: ACTVT Activity

Full authorization

Activities

S.	A...	Text
☐	02	Change
☒	03	Display
☐	BD	Maintain obj. in non-OwnerSys.

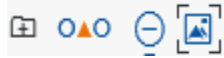
- Organizational field values are maintained in **organizational levels** option in authorization tab, whereas Normal field values can be **directly** maintained.

Field Vals of OrgLevels

Org. Level	'From'	'To'	More...	Change
☐ Plan Version				

Difference between Normal field and organizational field

Normal field	organizational field
*Does not represent organization Ex = > Activity.	*Represent organization Ex = > plant, company code.
*Field value meaning is universal ex = > Activity 03 means display across the world.	*Org field value meaning differs from organization to organization means specific to organization.
*Field values are maintained locally.	*Org field values have to be maintained centrally from the organization tab.



Note => When we click on this symbol called "**Where Used List**" (two hills with the sun rising in the background), it will show for which T-code the authorization object is used and also shows auth fields and its value range.

Role X:BS:ABAP_DEVELOPMENT:TATA

Transactions, RFC functions, and services that require authorization object S_DATASET:

Type of Application	Applicati...	Short Description	Authorization fld	Value range
Transaction	SE38	ABAP Editor	ACTVT	33
				34
			FILENAME	
			PROGRAM	SAPLOCAL_EDT1

Statuses of authorization objects

- There are **four** types.
- * **Standard** => Represent SAP default **standard values** pulled from SU24 for the T codes added in the **Menu**.

Standard	Old	Authorization for file access	S_DATASET
Standard	Old	Authorization for file access	T-W875039000

- * **Maintained** => Represents **blank fields** have been maintained with **values**.

Before ↘

Standard	Old	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
Standard	Old	Table Maintenance (using standard tools such as SM30)	T-W875039000
Activity		03	ACTVT
Table Authorization Group			DICBERCLS

- * After, fill the blank field with **some values** the status has changed to maintained. ↘

00	Maintained Old	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
00	Maintained Old	Table Maintenance (using standard tools such as SM30)	T-W875039000
*	Activity	03	ACTVT
*	Table Authorization Group	12	DICBERCLS

* **Changed** => Represent SAP **default values** have been maintained.

Before ~

00	Maintained Old	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
00	Maintained Old	Table Maintenance (using standard tools such as SM30)	T-W875039000
*	Activity	03	ACTVT
*	Table Authorization Group	12	DICBERCLS

* Maintain **some value** in Normal field (**Activity**) and click on save, the status will be updated called **changed**.

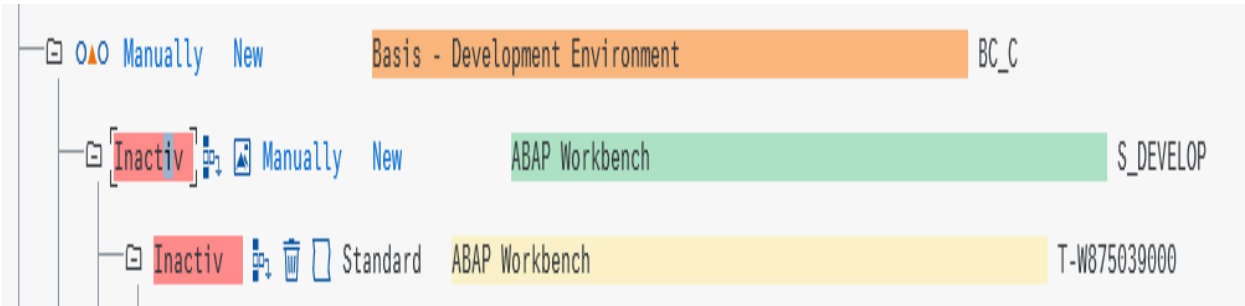
0	Changed New	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
00	Changed Old	Table Maintenance (using standard tools such as SM30)	T-W875039000
*	Activity	02, 03	ACTVT
*	Table Authorization Group	12	DICBERCLS

* **Manually** => Represent **authorization object** have been inserted directly into the role, not pulled from **SU24**.

0	Manually New	Basis - Development Environment	BC_C
0	Manually New	ABAP Workbench	S_DEVELOP

Deleting authorization object from role

- To **delete** authorization objects below two conditions, have to be satisfied: -
 - ✓ Authorization object should be **deactivated**.

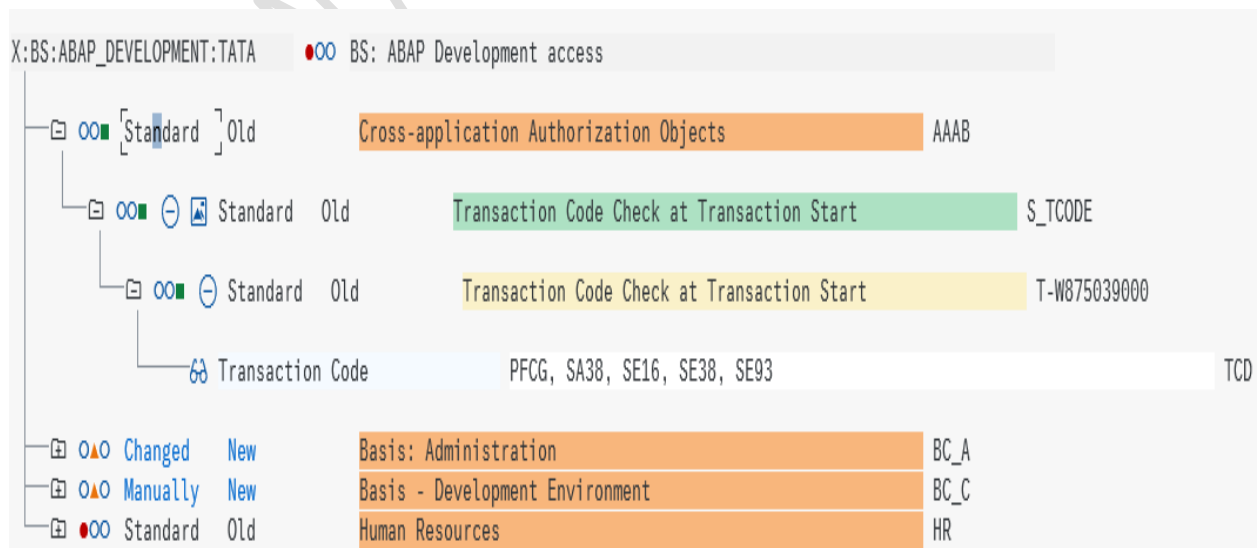


- ✓ Authorization object **status** should be either **manually** or **changed**.

Options in Authorization Tab Screen



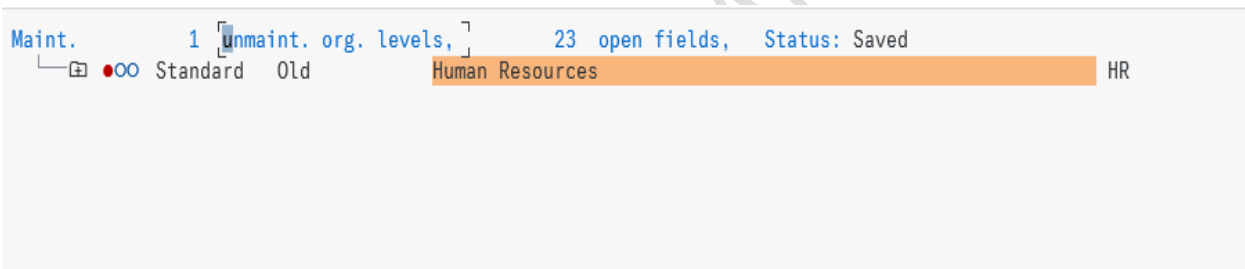
- ❖ **Save** => when we click on **save** all the data will be saved.
- ❖ **Expand** => When we click on a **particular field** and select the expand option, the entire field will be expanded.



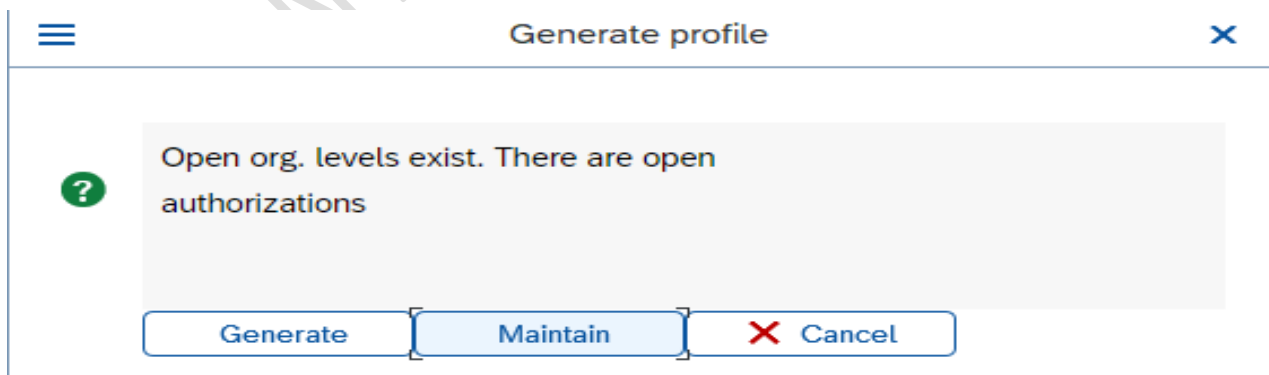
- ❖ **Collapse** => When some fields are expanded, clicking on the **collapse** option will collapse that particular entry.



- ❖ **Position** => When you click on any **particular field** and then select the position option, the system will take you to the **top** of the screen. After scrolling up, it will return to the **normal view**.



- ❖ **Profile** = After making changes to **authorizations** in the role, click on the profile option to **generate**. You will be prompted with two options, check, and click on generate.



- ❖ **Delete inactive** => To delete any authorizations, click on the specific field and select the **delete** option.
- ❖ Only **manually** added and **changed** authorizations will be deleted.



Delete Inactive Authorizations



Only modified and manual authorizations are deleted. Authorizations with the status "Standard" and "Maintained" are kept.



- ❖ **Selection criteria** => When you select this option, a list of **classes** will be displayed. Extend one class and select "**Insert Chosen**" above. This will add the authorization object and show its status as "**manually**."

< **SAP** Change Role: Insert Authorizations

✓ Insert chosen Cancel

Choose the authorizations you want to insert

All authorizations sorted by class

- AGRI: Authorization Objects /AGG
- Object class for Ariba /ARB
 - Authorization Object for Ariba Class Objects /ARBA/CLAS
 - Business Document Service: Cla CLASSNAME
 - Business Document Service: Cla CLASSTYPE
 - Authorization object for program execution /ARBA/PROG
 - Authorization object for RFC /ARBA/RFC
- Commercial Project Management /CPD
- SAP F15 Schnittstelle /F15
- GEO.e Authorizations /GEO

- ❖ **Manually** => Here, you can **directly** insert authorization objects into the role, without pulling them from SU24 or any of the T-codes. Type any authorization object and click "**Continue**."

0 unmaint. org.

S:ABAP_DEVELOPMENT:TATA

Standard Cross-app
Manually Basis: Ad
Manually Basis - De
Standard Human Res

Manual selection of authorizations

Authorization Object	S_Transprt	
Authorization Object	S_dataset	
Authorization Object		
Authorization Object		
Authorization Object		
Authorization Object		
Authorization Object		
Authorization Object		

AA
BC
BC
HR

✓ ✗

❖ **Open Authorizations** => When you click on "open authorizations" all empty fields will be displayed. However, SAP recommends that open fields should be **zero**.

Maint. 0 unmaint. org. levels, 25 open fields, Status: Changed

X:BS:ABAP_DEVELOPMENT:TATA BS: ABAP Development access

Standard Cross-application Authorization Objects AAAB
Manually Basis: Administration BC_A

Manually Authorization for file access S_DATASET

Standard Authorization for file access T-W875039001

Activity	33, 34	ACTVT
Physical file name		FILENAME
Program Name with Search Help	SAPLOCAL_EDT1	PROGRAM

Changed Authorization for file access T-W875039000
Manually Authorization for file access T-W875039002

Activity		ACTVT
Physical file name		FILENAME
Program Name with Search Help		PROGRAM

Changed Table Maintenance (using standard tools such as SM30) S_TABU_DIS

Standard Table Maintenance (using standard tools such as SM30) T-W875039001

Activity	03	ACTVT
Table Authorization Group		DICBERCLS

- ❖ **Changed** => When we click on this, it will **expand** to show how many objects have **changed status**.

Maint. 0 unmaint. org. levels, 25 open fields, Status: Changed

X:BS:ABAP_DEVELOPMENT:TATA BS: ABAP Development access

- Standard Cross-application Authorization Objects AAAB
- Manually Basis: Administration BC_A
- Manually Authorization for file access S_DATASET
 - Standard Authorization for file access T-W875039001
 - Activity 33, 34 ACTVT
 - Physical file name FILENAME
 - Program Name with Search Help SAPLOCAL_EDT1 PROGRAM
 - Changed Authorization for file access T-W875039000
 - Activity 33, 34, A6 ACTVT
 - Physical file name 12 FILENAME
 - Program Name with Search Help SAPLOCAL_EDT1 PROGRAM
 - Manually Authorization for file access T-W875039002
 - Activity ACTVT
 - Physical file name FILENAME
 - Program Name with Search Help PROGRAM
- Changed Table Maintenance (using standard tools such as SM30) S_TABU_DIS

- ❖ **More** => when we click on this it will show, **many** options to maintain authorizations and **org levels**.

Maint. 0 unmaint. org. levels, 25 open fields, Status: Changed

X:BS:ABAP_DEVELOPMENT:TATA BS: ABAP Development access

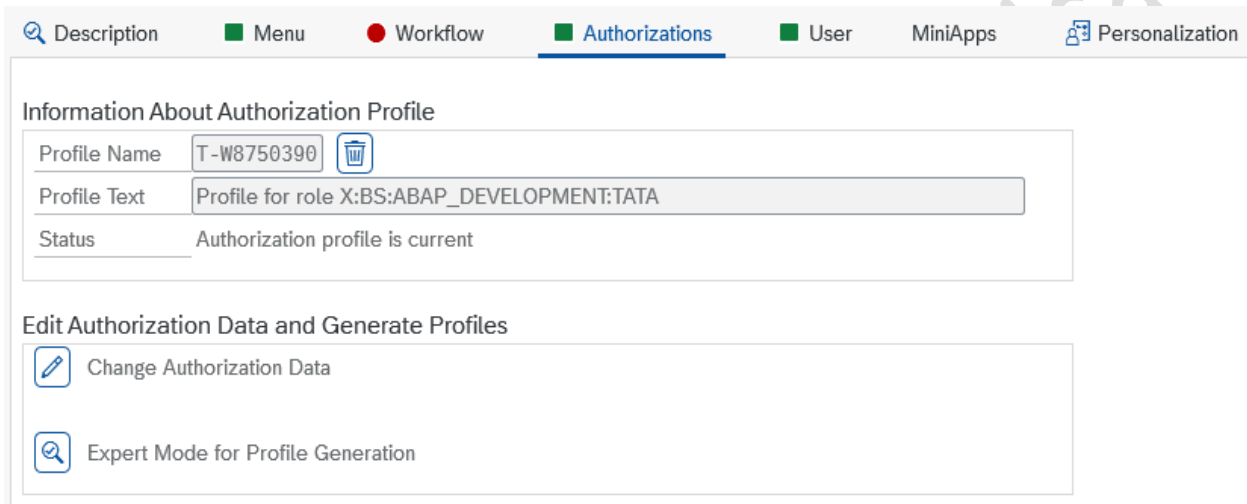
	Standard	Cross-application Authorization Objects	AAAB
	Manually	Basis: Administration	BC_A
	Manually	Basis - Development Environment	BC_C
	Standard	Human Resources	HR

More ▾
 Maintained Authorizations (Ctrl+F9)
 Organizational levels... (Ctrl+F8)
 Legend (Ctrl+Shift+F8)
 Information (Ctrl+F1)
 Evaluate Trace Data (Ctrl+F5)
 Versions (Ctrl+F7)
 Cancel (F12)
 First Page (Ctrl+Page up)
 Previous Page (Page up)
 Next Page (Page down)
 Last Page (Ctrl+Page down)

Profile Generation

✧ This is done in two ways in **Authorizations** tab.

- **Manual mode** => Change in **Authorization** data.
- **Expert mode** => This mode is used during **role modification** only, not during the role creation.



Information About Authorization Profile

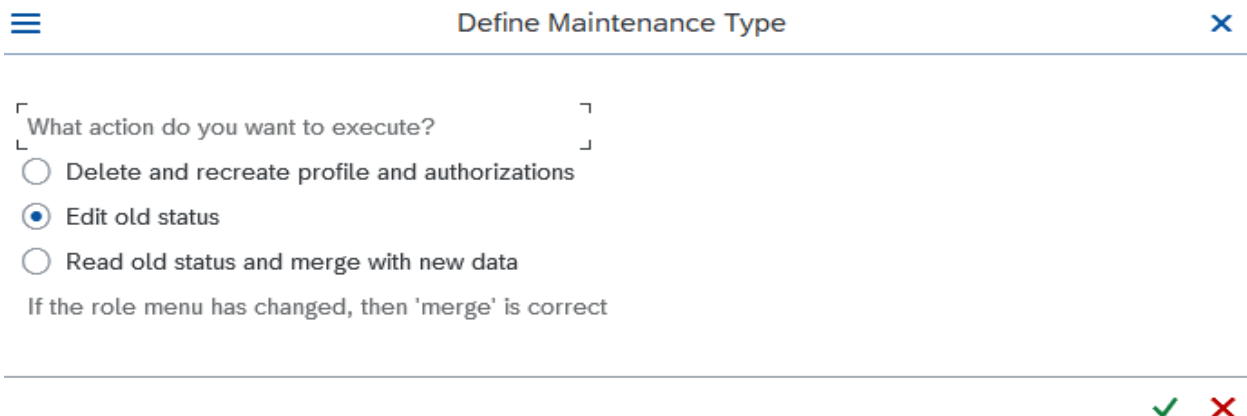
Profile Name	T-W8750390	
Profile Text	Profile for role X:BS:ABAP_DEVELOPMENT:TATA	
Status	Authorization profile is current	

Edit Authorization Data and Generate Profiles

- Change Authorization Data
- Expert Mode for Profile Generation

When we click on **expert mode for profile generation** there will be **three** actions need to be performed: -

- ⇒ Delete and recreate profile and authorizations.
- ⇒ Edit old status.
- ⇒ Read old status and merge with new data.



Define Maintenance Type

What action do you want to execute?

☐ Delete and recreate profile and authorizations

☒ Edit old status

☐ Read old status and merge with new data

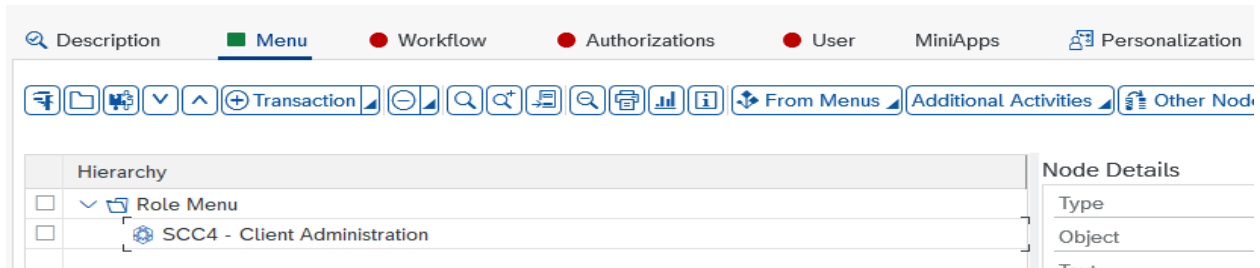
If the role menu has changed, then 'merge' is correct

✓ ✗

* Click on continue.

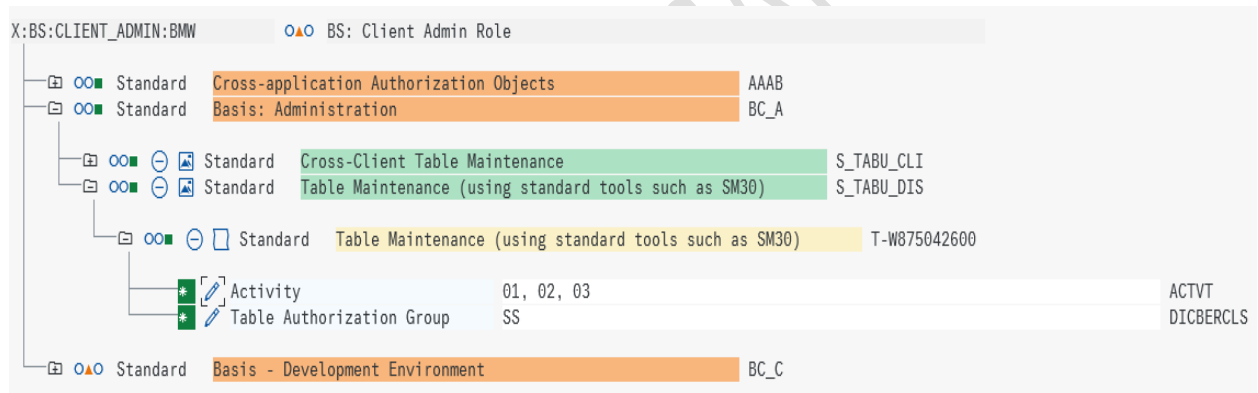
1) Delete and recreate profile and authorizations.

- Create a new role and add the T-code SCC4 in the Menu tab, then go to change the **authorization data** in the authorizations tab.

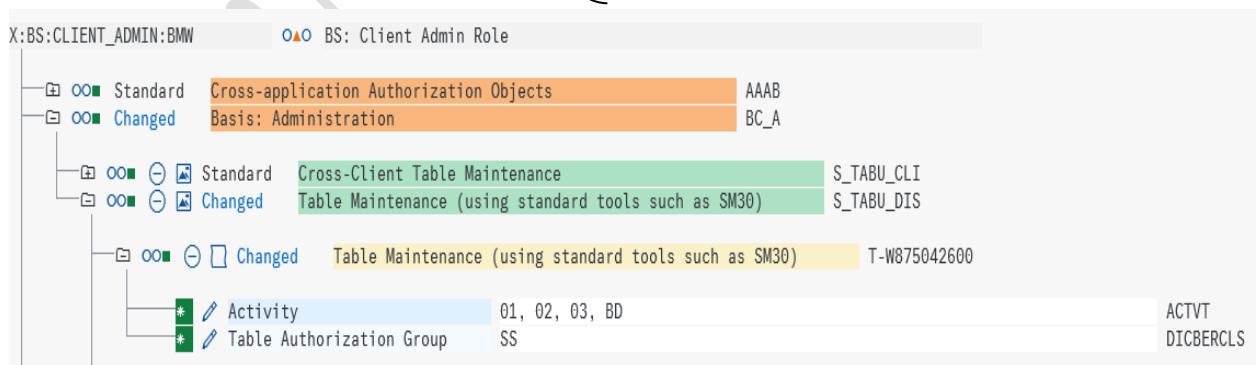


- In the **auth tab** screen the authorization object called **S_TABU_DIS** (any object), do Some customization, here I added Normal field (**Activity**) called **BD**.

Before



After



- Click on save and generate the Profile.

*Then Modify **the role**, go back, and add the T code called **SCC5** in the menu Tab.

Hierarchy

- ☐ Role Menu
 - ☒ SCC5 - Delete Client
 - ☐ SCC4 - Client Administration

Node Details

Type
Object
Text

- Go to the Authorizations tab. Instead of changing the authorization data, click on Expert Mode and select the first option, **"Delete and Recreate Profile and Authorizations."**

Information About Authorization Profile

Profile Name	T-W8750426	
Profile Text	Profile for role X:BS:CLIENT_ADMIN:BMW	
Status	Authorization profile is current	

Edit Authorization Data and Generate Profiles

 [Change Authorization Data](#)

 [Expert Mode for Profile Generation](#)

Check Now =>

- ✓ If the customization is to be deleted - old data is lost.
- ✓ It deletes entire **old data** including its **customization also**.
- ✓ Freshly loads the authorization data from **SU24 for the T codes** added in the **menu**.

X:BS:CLIENT_ADMIN:BMW		BS: Client Admin Role	
Standard	Cross-application Authorization Objects	AAAB	
Standard	Transaction Code Check at Transaction Start	S_TCODE	
Standard	Transaction Code Check at Transaction Start	T-W875042600	
Transaction Code	SCC4, SCC5		TCD
Standard	Basis: Administration	BC_A	
Standard	Data Import for Client Copy	S_CLNT_IMP	
Standard	Authorization for file access	S_DATASET	
Standard	Cross-Client Table Maintenance	S_TABU_CLI	
Standard	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS	
Standard	Table Maintenance (using standard tools such as SM30)	T-W875042600	
Activity	01, 02, 03		ACTVT
Table Authorization Group	SS		DICBERCLS
Standard	Authorizations: Role Check	S_USER_AGR	
Standard	User Master Maintenance: User Groups	S_USER_GRP	
Standard	User Master Maintenance: Authorization Profile	S_USER_PRO	
Standard	Basis - Development Environment	BC_C	

2) Edit old status.

Continue----

→ In the above same screen only, I will do some **customization** again, I will add the Normal field (**Activity**) called **BD**.

→ Click on save and **generate** the profile.

X:BS:CLIENT_ADMIN:BMW BS: Client Admin Role

Standard	Changed	Object	Role
Standard		Cross-application Authorization Objects	AAAB
Standard		Basis: Administration	BC_A
Standard		Data Import for Client Copy	S_CLNT_IMP
Standard		Authorization for file access	S_DATASET
Standard		Cross-Client Table Maintenance	S_TABU_CLI
Standard		Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
Standard	Changed	Table Maintenance (using standard tools such as SM30)	T-W875042600
	Activity	01, 02, 03, BD	ACTVT
	Table Authorization Group	SS	DICBERCLS
Inactiv	Standard	Table Maintenance (using standard tools such as SM30)	T-W875042601
Inactiv	Activity	01, 02, 03	ACTVT
Inactiv	Table Authorization Group	SS	DICBERCLS
Standard		Authorizations: Role Check	S_USER_AGR
Standard		User Master Maintenance: User Groups	S_USER_GRP
Standard		User Master Maintenance: Authorization Profile	S_USER_PRO
Standard		Basis - Development Environment	BC_C

→ Come back and add the **T code** called **SM36** in the Menu tab, save.

Description **Menu** Workflow Authorizations User MiniApps Personalization

Transaction From Menus Additional Activities Other Node Details

Hierarchy	Node Details
Role Menu	Type: T
SCC5 - Delete Client	Object: SI
SM36 - Schedule Background Job	Text: S
SCC4 - Client Administration	

→ Afte, go to the **Authorizations** tab and click on Expert Mode for profile generation. Then, select the second option called **"Edit Old Status."**

Define Maintenance Type

What action do you want to execute?

☐ Delete and recreate profile and authorizations

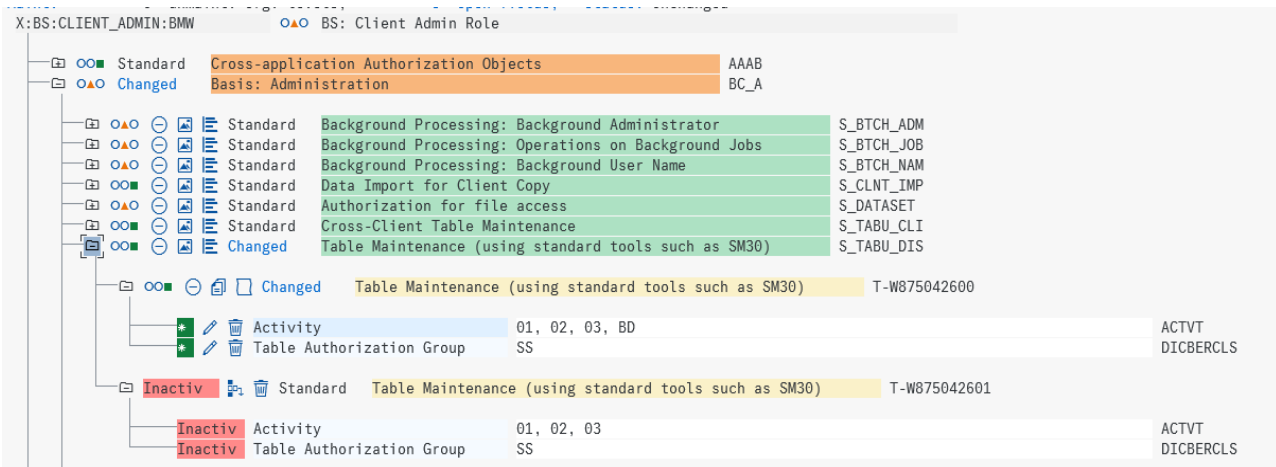
☒ Edit old status

☐ Read old status and merge with new data

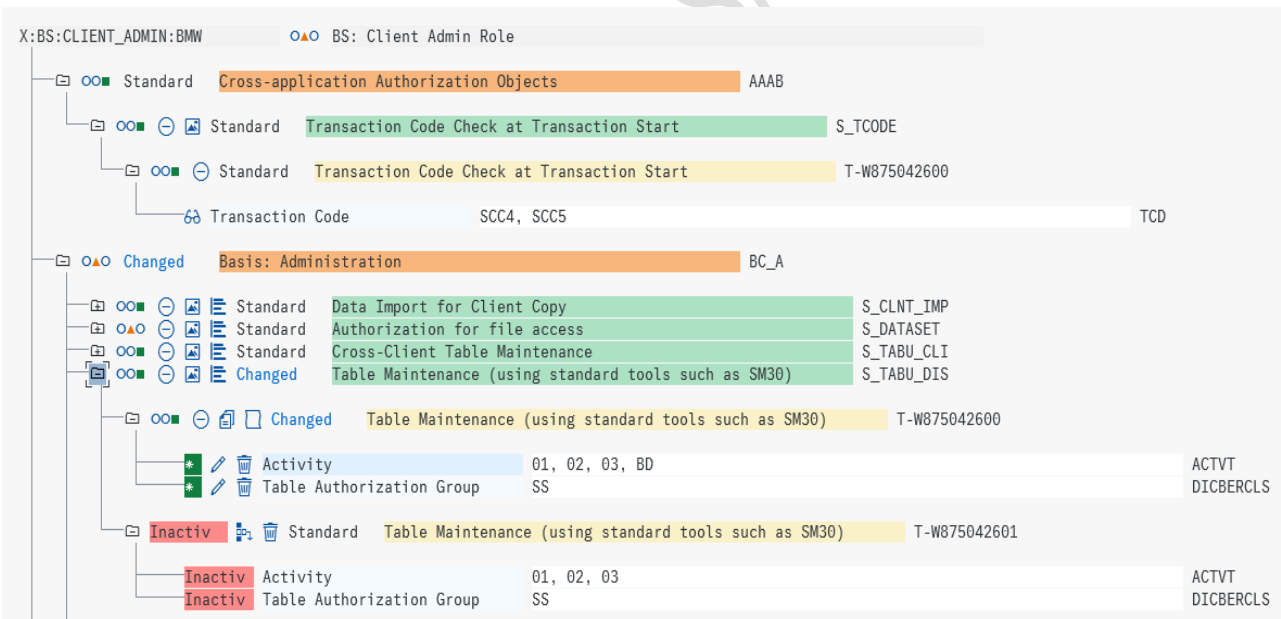
If the role menu has changed, then 'merge' is correct

→ If the authorization data has to be modified directly without modifying menu.

→ It keeps the **old data** as it is (including customization) also.



→ It does **not pull** the authorization data related to new **T codes** added in the menu.



3) Read old status and merge with new data.

Continue----

- Come back and go to the **Authorizations tab**. Click on Expert Mode for profile generation, then select the third option called **"Read Old Status and Merge with New Data."**



Define Maintenance Type



What action do you want to execute?

- ☐ Delete and recreate profile and authorizations
- ☐ Edit old status
- ☒ Read old status and merge with new data

If the role menu has changed, then 'merge' is correct

➤ It keeps the **old data** as it is (including customization) also.

X:BS:CLIENT_ADMIN:BMW BS: Client Admin Role

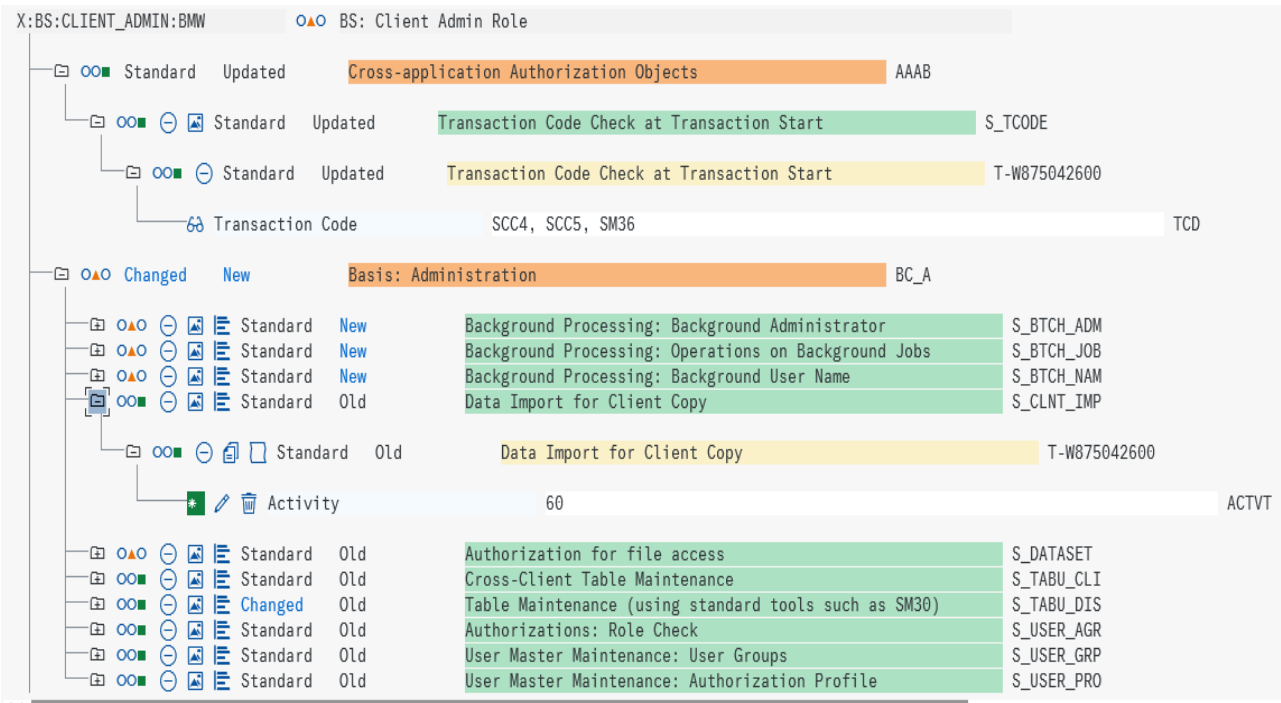
Standard	Updated	Cross-application Authorization Objects	AAAB
Changed	New	Basis: Administration	BC_A
Standard	New	Background Processing: Background Administrator	S_BTCH_ADM
Standard	New	Background Processing: Operations on Background Jobs	S_BTCH_JOB
Standard	New	Background Processing: Background User Name	S_BTCH_NAM
Standard	Old	Data Import for Client Copy	S_CLNT_IMP
Standard	Old	Authorization for file access	S_DATASET
Standard	Old	Cross-Client Table Maintenance	S_TABU_CLI
Standard	Old	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
Standard	Old	Table Maintenance (using standard tools such as SM30)	T-W875042600
Activity	01, 02, 03, BD	ACTVT	DICBERCLS
Table Authorization Group	SS		
Inactiv	Standard	Table Maintenance (using standard tools such as SM30)	T-W875042601
Inactiv	Activity	01, 02, 03	ACTVT
Inactiv	Table Authorization Group	SS	DICBERCLS
Standard	Old	Authorizations: Role Check	S_USER_AGR
Standard	Old	User Master Maintenance: User Groups	S_USER_GRP
Standard	Old	User Master Maintenance: Authorization Profile	S_USER_PRO

➤ It also pulls the new authorization data related to all the **T codes** added in the menu.

X:BS:CLIENT_ADMIN:BMW BS: Client Admin Role

Standard	Updated	Cross-application Authorization Objects	AAAB
Standard	Updated	Transaction Code Check at Transaction Start	S_TCODE
Standard	Updated	Transaction Code Check at Transaction Start	T-W875042600
Transaction Code	SCC4, SCC5, SM36	TCD	
Changed	New	Basis: Administration	BC_A
Standard	New	Background Processing: Background Administrator	S_BTCH_ADM
Standard	New	Background Processing: Operations on Background Jobs	S_BTCH_JOB
Standard	New	Background Processing: Background User Name	S_BTCH_NAM
Standard	Old	Data Import for Client Copy	S_CLNT_IMP
Standard	Old	Authorization for file access	S_DATASET
Standard	Old	Cross-Client Table Maintenance	S_TABU_CLI
Standard	Old	Table Maintenance (using standard tools such as SM30)	S_TABU_DIS
Standard	Old	Authorizations: Role Check	S_USER_AGR
Standard	Old	User Master Maintenance: User Groups	S_USER_GRP
Standard	Old	User Master Maintenance: Authorization Profile	S_USER_PRO
Standard	Old	Basis - Development Environment	BC_C

- It also merges the **old data** with the **new data** wherever is possible.
- Without losing the **old data** merge the new data of the **T code** inserted in the menu.



Note: -

- ✧ Out of these three options the most favorable option is **Read old status and merge with new data**.
- ✧ During role creation there is no something called as **old data** hence we do not go for expert mode in the **role creation** concept.

Composite Role

- It is a group of single roles.

Scenario: -

- ✓ Two abap roles =>

***Abap display single role.**

***Abap full access single role.**

✓ Two roles for basis team =>

***Basis display role.**

***Basis admin role.**

✓ Two security roles =>

***Security display role**

***Security admin role**

Security team needs below role.

✧ **Security admin role.**

✧ **Abap display access.**

- ❖ These three single roles are combined together to create **security composite role**.
- ❖ It will be easier to select the composite role because it is **minimum in number**.
- ❖ Note => The most **important problem** is when user is requesting the role, instead of requesting display role, sometimes the user will be requesting **full access role**, then that request will be rejected, because the approver, who is approving the request he will see the different user is requesting full access, so he may **reject** the request.
- ❖ The key role of composite role means to make the **user comfortable** and also it **decreases** the security admin task.

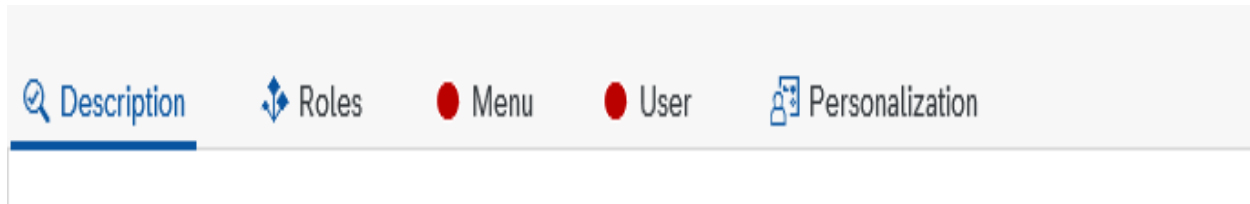
Composite role Tabs

- ✧ Enter a T code called **PFCG** in sap command field.
- ✧ Type the Composite role name according to the **role naming convention**.
- ✧ Instead of Creating single role click on **composite role Option**.

The screenshot shows the SAP Role Maintenance interface. At the top, there is a header bar with the SAP logo and the title "Role Maintenance". Below the header, there is a toolbar with various icons including a checkmark, a dropdown menu, a copy icon, a delete icon, a refresh icon, an information icon, and buttons for "Transactions" and "Cancel". The main area contains a form with the following fields and controls:

- Role:** A text field containing the value "Y:BASIS_COMPOSITE_ROLE". To the right of this field are icons for a document, a pencil, and a magnifying glass.
- Short Description:** An empty text field.
- Buttons:** Below the "Role" field, there are two buttons: "Single Role" and "Comp. Role". To the right of these buttons are icons for a star, a document, and a group of people.
- Views:** At the bottom left, there is a section labeled "Views" with icons for a list, a table, a chart, and a refresh button, followed by a "Show Documentation" button.
- Favorites:** At the bottom, there is a section labeled "Favorites" with a list of roles.

- ⇒ When we click on **composite role** It will display the below screen with some tabs and also You can see the difference between **single and composite role tabs**.



- ⇒ **Description** => just fill the long text with the information like **why**, when and **who** is creating the role.

Administration Information

Created	
User	VARUN
Date	26.07.2024
Time	11:04:13

Long Text

26/07/24 cr1234 varun Basis composite role

- ⇒ **Roles** => Fill the **single roles**.

Role	Name	Target Sys	Act.
☐ X:BS:ABAP_DEVELOPMENT:TATA	BS: ABAP Development access	Own System	☒
☐ X:BS:CLIENT_ADMIN:BMW	Client Admin Role	Own System	☒
☐			☐
☐			☐

- ⇒ **Menu** => Click on **import menu** to see the single roles and **t code** in it.
- ⇒ To convert menu to **green** color click on **import menu** option.

Description Roles **Menu** User Personalization

Import Menu Additional Activities Other Node Details Menu Options

Hierarchy

- Role Menu
 - X:BS:ABAP_DEVELOPMENT:TATA
 - PFCG - Role Maintenance
 - SE38 - ABAP Editor
 - SA38 - ABAP Reporting
 - SE16 - Data Browser
 - SE93 - Maintain Transaction Codes
 - X:BS:CLIENT_ADMIN:BMW

Node Details

Type
Object
Text

⇒ **User Tab** => indicates the **list of users** and assigned to the role, click on user comparison, once done you will see the user comparison tab become **green**.

Description Roles Menu **User** Personalization

Selection Organizational Mg... User Comparison

User Assignments

User ID	User Name	From	to	I...

⇒ **Personalization tab** => Do not touch it.

Description Roles Menu User **Personalization**

Description Personalization object key

WorkDeck Personalization: Data Provider	/GBFRM/MYV_PERS_DATAPROVIDER
Personalization in Issue and Activity Mana...	/IAM/PERS
Line Maintenance - Aircraft Accident: Use...	/ISDFPS/LM_ACC
Line Maintenance - Flight: User Settings	/ISDFPS/LM_FLIGHT

Note => When we assign the **composite role** to the any user in the user roles tab another tab will be getting called **indirect assignment tab**, we can directly **delete**

the composite role which was assigned, but not able to **delete directly single roles** in that.

User	TESTID		
Changed By	VARUN	26.07.2024	11:21:34
Status	Revised		

Documentation Address Logon Data SNC Defaults Parameters Rules Profiles Groups Personalization Lic. Data

Reference User

Role Assignments

	Status	Role	Ty...	Start Date	End Date	Short Role Description	Indirect Assignment
☐		X:BS:ABAP_DEVELOPMENT:TATA		26.07.2024	31.12.9999	BS: ABAP Developme...	
☐		Y:BASIS_COMPOSITE_ROLE		26.07.2024	31.12.9999	Y: Basis Composite ro...	=
☐		X:BS:CLIENT_ADMIN:BMW		26.07.2024	31.12.9999	BS: Client Admin Role	

Master Role

- It is a **parent role** which has **full access** and contains **org level** as *

Derived role => It is a **single role** which is **derived** from another **single role**.

=== What is the **necessity** of a derived role **means**?

When the **client** is

=> MNC - coke, BMW, nestle

=> Multiple business company – Tata, Dupont, total, Unilever

Coke

India finance => process invoices of India.

USA finance => process invoices of USA.

T code used to process invoice is same FBV0.

In India when they will execute the FBV0, they should be able to process the **invoices of India only.**

In USA when they will execute the FBV0, they should be able to process the **invoices of USA only**.

This **access** can be differentiated with the **org fields** i.e., plant, cost center, company code.

In these diagram **three roles** are derived from **one Master role**.

Role Finance

T code => FBV0

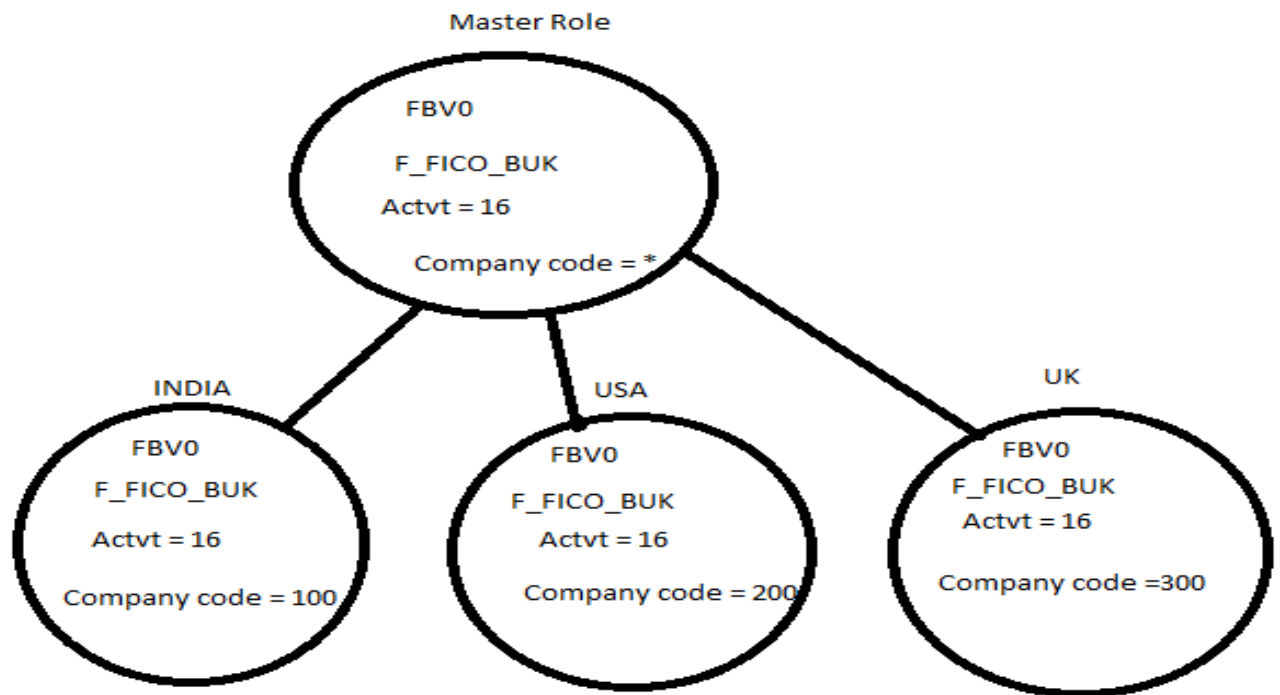
Auth object => F_FICO_BUK

Activity = 16

Company code = *

For the **Maintenance** become **easier** we will use this **concept**.

- If any **user requests extra access** like t code or auth object, instead of modifying **derived roles** we can directly do modifying in **Master role**.



- When You derive the roles from Master roles the **T code, authorization objects** have been derived as it is.
- The Only difference between Master and derived role is the **org field Values**.

Here, I created a **Master role** with the t-codes PFCG, RSA1, and RSD1 and maintained the **org value** Plan version as *. Additionally, I customized the **S_USER_AGR** object with the role name as **X*** and generated the **profile**.

And also, I created another **derived role** for India. In the Description tab, there is an option called "**Derived from Role.**" In that field, I specified the **Master role**. This is an extra thing you **need** to do in a **derived role**.

And also, we got a option called delete **Inheritance relationship** that means the **relationship** has been established between **Master** and **derived role**.

Role

Role: Z:X:BW:REPORTING_ACCESS_INDIA ☐ Obsolete

Description: derived role for India

Target System: ☐ No destination

Administration Information

	Created	Changed
User	VARUN	VARUN
Date	26.07.2024	26.07.2024
Time	15:24:25	15:24:50

Transaction Inheritance

Derive from Role: X:BW:REPORTING_ACCESS

X: Bw report access

[Delete Inheritance Relationship](#)

We can see the **T codes** whatever there in the **Master role** it will pull to the **derived role**., and also here I maintained **org value** Plan version as **01**.

Menu

Hierarchy

- ☐ Role Menu
 - ☐ RSA1 - Modeling - DW Workbench
 - ☐ RSD1 - Configure Characteristics
 - ☐ PFCG - Role Maintenance

Node Details

Type

Object

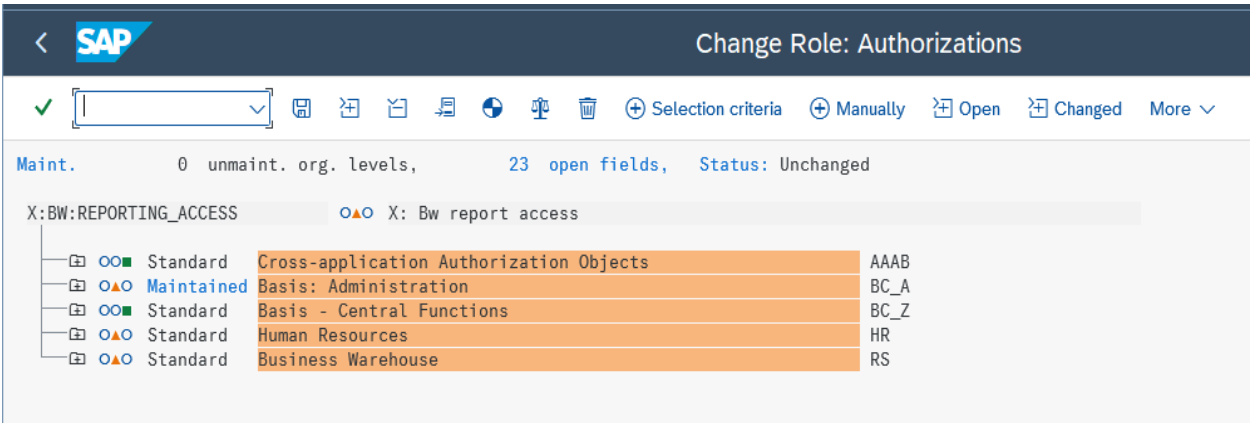
Text

Whatever I done **customization** in Master role It will **not** pull to this **role**, in order to pull that **customization**, again go back to **master role** there will be another tab generated called **generate derived role**, click on this.

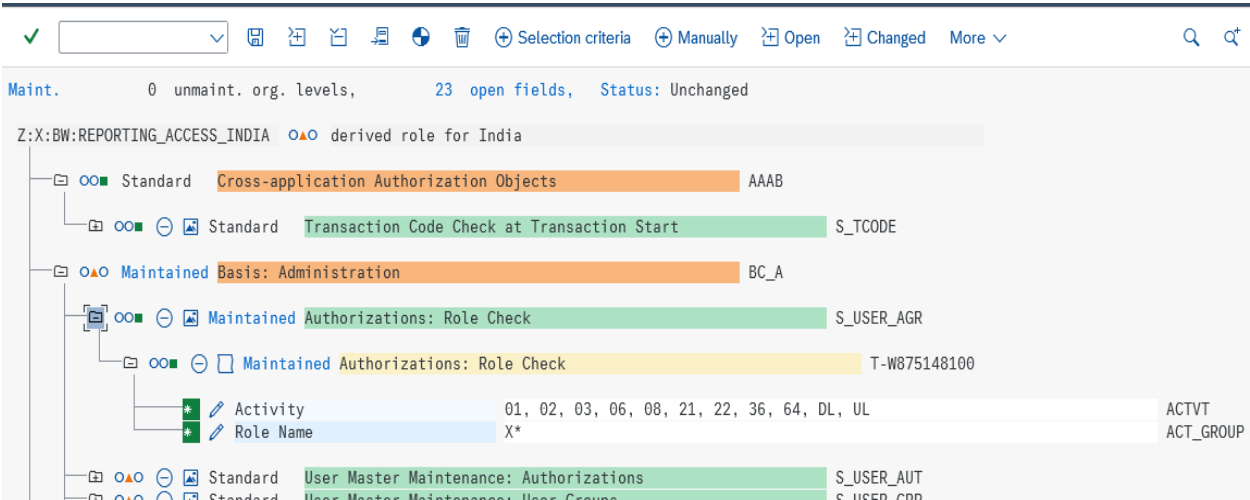
Maint. 0 unmaint. org. levels, 24 open fields, Status: Changed

Z:X:BW:REPORTING_ACCESS_INDIA ☐ derived role for India

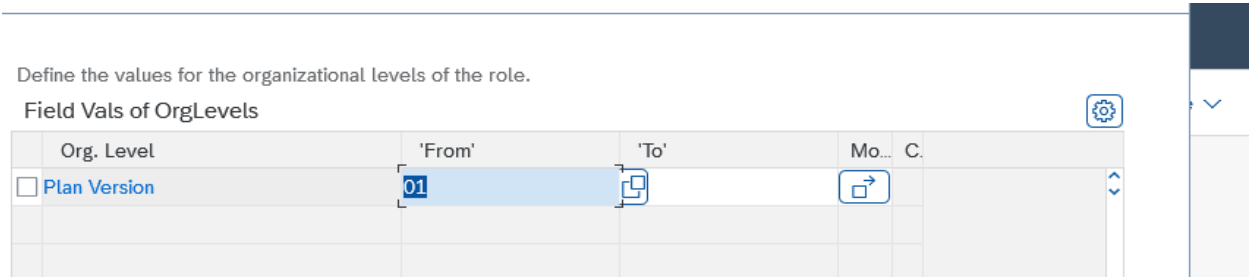
- ☐ Standard Cross-application Authorization Objects AAAB
 - ☐ Standard Transaction Code Check at Transaction Start S_TCODE
- ☐ Standard Basis: Administration BC_A
 - ☐ Standard Authorizations: Role Check S_USER_AGR
 - ☐ Standard Authorizations: Role Check T-W875148100
 - ☐ Activity 01, 02, 03, 06, 08, 21, 22, 36, 64, DL, UL ACTVT
 - ☐ Role Name ACT_GROUP
 - ☐ Standard User Master Maintenance: Authorizations S_USER_AUT



After clicking **Generate derived roles** tab, now go **back and check** in the derived role **everything** the data will pushed to **derived role** expect **org values**.



But we see the **Org Values** in the tab whatever we **maintained** in derived role it will not change.so that **org value** is **not overwritten**, only the **normal fields** are **overwritten**.



✚ **Another Example** => Create a **role** with **company code** value as * and we have derived role for Germany, Malaysia, UK, with **company code** value as **200,300,400**. Suppose if we add a **new t code** to master role and if we create derive role from it, it will push the t code to Germany, Malaysia, UK, role, but it will **not touch** the company code Value of **derived roles**. The **company code** value will remain as it is. **That is the concept of Master and derived role.**

If you want to add a **new t code FB01** in these **three** derived roles, then you need to add this **t code** to **master role** and derived roles will get **modified** accordingly.

The only difference between **Master** and **derived role** is **org field values**. i.e., **company code** values in this example.

Question => How many **derived roles** can be derived in **one Master role**? Is there any limit?

✧ **No limit** it depends on **How many branches** or business the **company has**.

Role Tables

All the Tables related to **roles** starts with **AGR*** (Activity Group => Role Name)

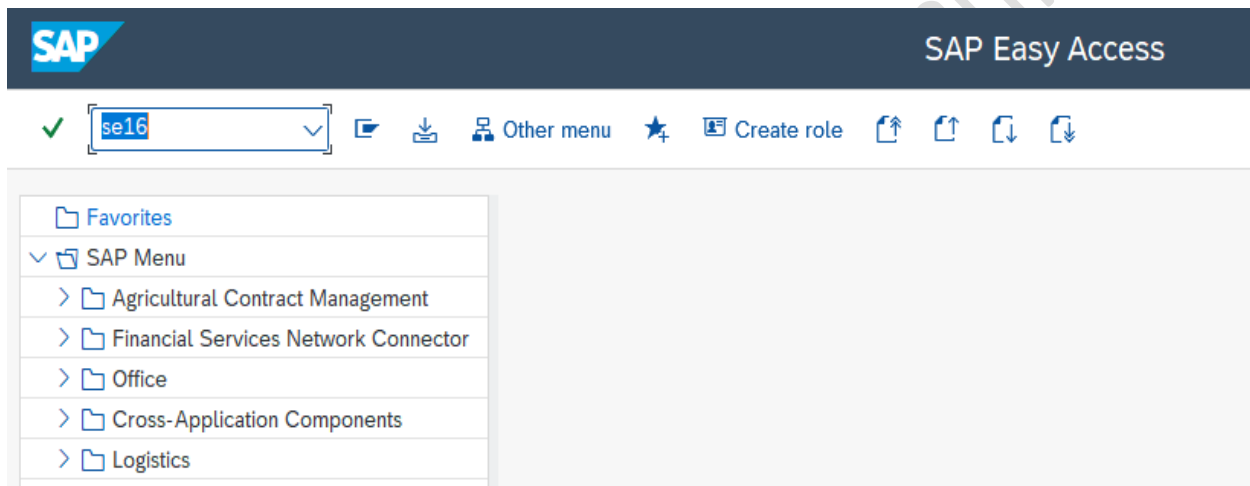
Some **Important Tables** are: -

- **AGR_DEFINE** => All roles definition and Master, derived role information.
- **AGR_AGRS** => Single and Composite roles Information
- **AGR_TCODES** => Roles and its PFCG Menu's Transactions.
- **AGR_USERS** => User assigned to a Role.
- **AGR_1250** => Authorization data for the activity group
- **AGR_1251** => Role and its Authorization Values.

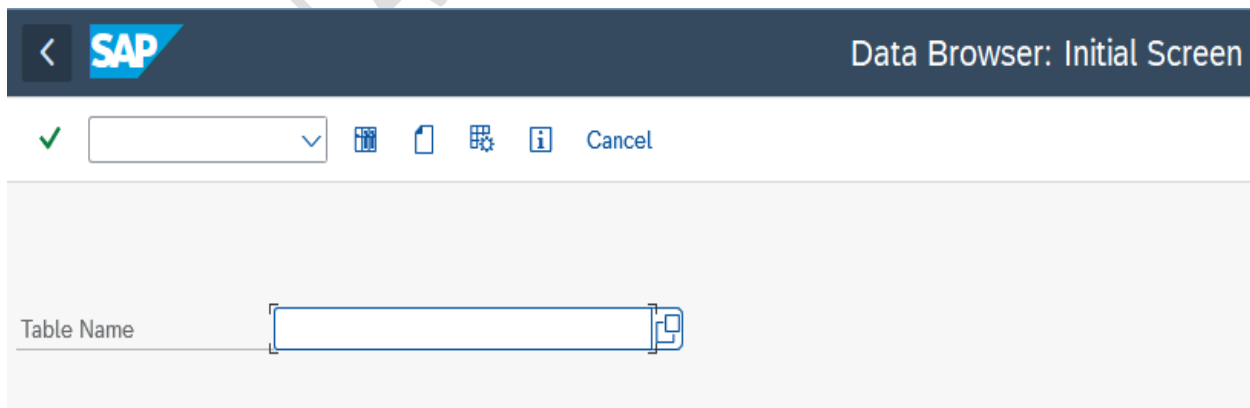
- **AGR_1252** => Role and its Org Values.
- **AGR_INFO** => Filter Values from Generation Run.
- **AGR_TIME** => Time Stamp for Role (Menu, Profile, Authorizations)
- **AGR_NUM_2** => Internal Counter for Assigning Profile Names.
- **AGR_TIME** => Role Modification details with date and time.

How to **view** the tables means ----

- Enter a T code called **SE16** in a sap Command field and click on **enter**.



- The below **initial screen** will Open.



- For ex --- Type the Table called **AGR_DEFINE** in the Table Name and Click on **enter**.

✓

▼

Cancel

Table Name

agr_define

- After this, the below **screen** will open. Type the **role name** or use the **search** option. If needed, you can also **apply filters**.

< **SAP** Data Browser: Table AGR_DEFINE: Selection Screen

✓

AGR_NAME	Y_BASIS_D_APAC_DIS	v		v
PARENT_AGR		to		v
CREATE_USR		to		v
CREATE_DAT		to		v
CREATE_TIM	00:00:00	to	00:00:00	v
CREATE_TMP		to		v
CHANGE_USR		to		v
CHANGE_DAT		to		v
CHANGE_TIM	00:00:00	to	00:00:00	v
CHANGE_TMP		to		v
ATTRIBUTES		to		v

Width of Output List

Maximum No. of Hits

- Then you can click on **execute** to see what **AGR_DEFINE** Table contains.

SAP Data Browser: Table AGR_DEFINE Select Entries 1

Back (F3) [Search Icon] [Navigation Icons] [Check Table...] Cancel [Print Icon] [More v]

Table: AGR_DEFINE
Layed Fields: 12 of 13 Fixed Columns: [2] List Width: [0250]

ANDT	AGR_NAME	PARENT_AGR	CREATE_USR	CREATE_DAT	CREATE_TIM	CREATE_TMP	CHANGE_USR	CHANGE_DAT	CHANGE_TIM	CHANGE
12	Y_BASIS_D_APAC_DISP_VARUN	Y_BASIS_S_DISPLAY_VARUN	VARUN	01.07.2024	22:52:59	0	VARUN	01.07.2024	23:09:53	

---- This table contains all **role definitions**, including Master (parent) and derived **role information**. It also shows the user, change date, and time, which can be filtered.

For Ex, another table called **AGR_1252**--- see in the below picture it will contain **role name and its Org Values**.

Data Browser: Table AGR_1252 Select Entries

Table: AGR_1252
Displayed Fields: 6 of 6 Fixed Columns: [3] List Width: 0250

MANDT	AGR_NAME	COUNTER	VARBL	LOW	HIGH
☐ 812	Y:PM708-PROJECTS-PURCHASEOFR	000001	\$SEKGRP	*	
☐ 812	Y:PM708-PROJECTS-PURCHASEOFR	000002	\$SEKORG	*	
☐ 812	Y:PM708-PROJECTS-PURCHASEOFR	000003	\$SWERKS	*	
☐ 812	Y_BC_C_BASIS_USER	000001	\$PLVAR		
☐ 812	Y_BC_C_BASIS_USER	000002	\$VKORG		
☐ 812	Y_BC_C_BASIS_USER	000003	\$SWERKS		
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN	000001	\$PLVAR	*	
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN	000002	\$VKORG	*	
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN	000003	\$SWERKS	*	
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN1	000001	\$PLVAR	*	
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN1	000002	\$VKORG	*	
☐ 812	Y_BC_P_DISPLAY_BASIS_ADMIN1	000003	\$SWERKS	*	
☐ 812	Y_BC_S_DISPLAY_BASIS_ADMIN	000001	\$PLVAR	*	
☐ 812	Y_BC_S_DISPLAY_BASIS_ADMIN	000002	\$VKORG	*	
☐ 812	Y_BC_S_DISPLAY_BASIS_ADMIN	000003	\$SWERKS	*	
☐ 812	Y_BC_S_MAINT_BASIS_ADMIN	000001	\$PLVAR		
☐ 812	Y_BC_S_MAINT_BASIS_ADMIN	000002	\$VKORG		
☐ 812	Y_BC_S_MAINT_BASIS_ADMIN	000003	\$SWERKS		
☐ 812	Y_BC_S_SU53_TEST	000001	\$PLVAR	*	
☐ 812	Y_BC_S_SU53_TEST	000002	\$VKORG	*	
☐ 812	Y_BC_S_SU53_TEST	000003	\$SWERKS	*	
☐ 812	Y_EHS_D_APAC_MAINT_VARUN	000001	\$SWERKS	3500	
☐ 812	Y_EHS_D_APAC_MAINT_VARUN	000002	\$SWERKS	CAP2	
☐ 812	Y_EHS_D_APAC_MAINT_VARUN	000003	\$SWERKS	MKP1	

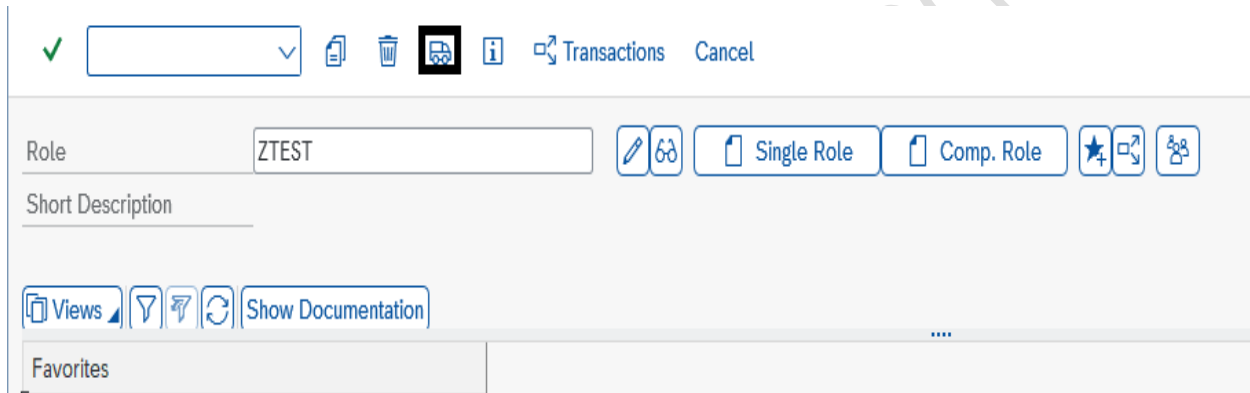
Role Deletion

= When the role exists in the **Development** environment only

- Go to **PFCG**.
- Put the **role name**.
- Delete the **role**.

= When the Role exists in Dev, quality, and PRD Systems, especially in **Quality and production**, we will have **PFCG** display access only.

- Go to **PFCG**
- Put the **role** name.
- Create **Transport request**.
- Load the **role** into transport request.
- Delete the **role** in development system.
- Release the **Transport request**.



✓ [dropdown] [icons] Transactions Cancel

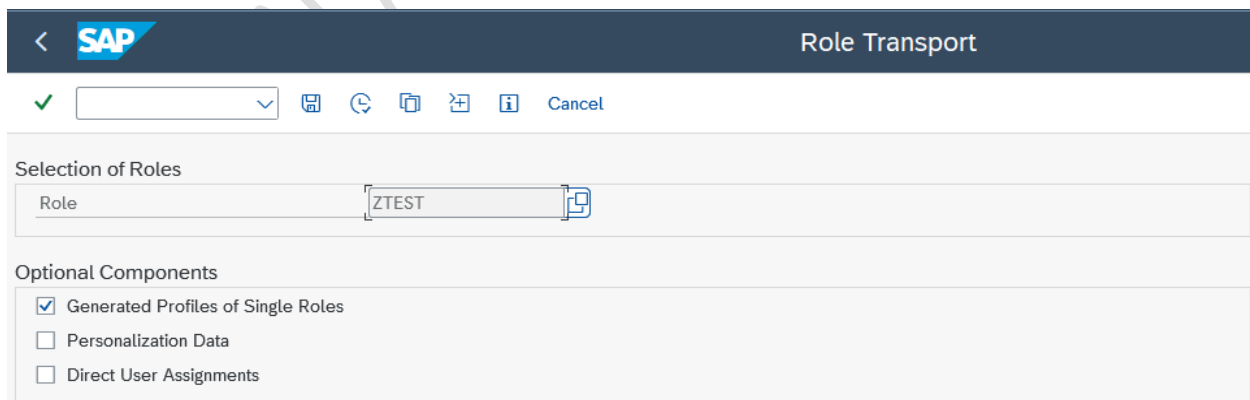
Role: ZTEST [edit] [copy] [Single Role] [Comp. Role] [star] [share] [help]

Short Description: _____

[Views] [filter] [refresh] [Show Documentation] ...

Favorites: _____

After clicking **Transport role** option, the below screen will come in these **never tick** Personalization data **and** direct user assignment.



< SAP Role Transport

✓ [dropdown] [icons] Cancel

Selection of Roles

Role: ZTEST [copy]

Optional Components

☒ Generated Profiles of Single Roles

☐ Personalization Data

☐ Direct User Assignments

After that click on **execute** and Create **Transport request**, save and continue too Next.

Create Request

Request: Customizing request

Short Description: Testing the role Cr123 varun

Project: [Empty]

Owner: VARUN

Status: New

Last changed: 29.07.2024 15:02:08

Source client: 812

Target: W53

Tasks:

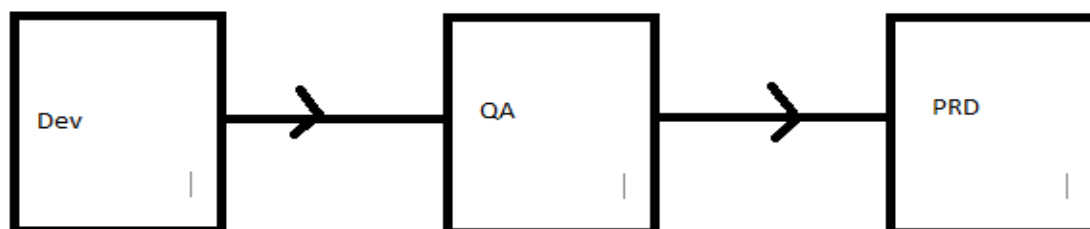
User
VARUN

After saving the **TR**, it shows the below screen which it contains **Request Number**, Role Type, and the role status, if it is in **green** means then the role recorded **successfully**.

Request W88K902287 contains all green roles						
Stat...	Role ▲	Type	Short T...	Single R...	Imparting R...	Error Messa...
oo	ZTEST					

TMS – Transport Management System

--**Transport** => Movement of **data** from one **environment** to another **environment**.



➤ Transport is done the form of **Transport request**.

- Transport request name => **SID K9 5-digit Request no.**
- Example => **W88K900035**
- **SE01, SE09, SE10** These **T codes** are used to create, display, release the Transport request.
- Transport Requests are of **4 Types** but mainly most of us use **only Two** They are: -

✓ [Dropdown] [Icons] Cancel

Display **Transports** Piece Lists Client Deliveries

User [VARUN] [Icon]

Request Type

- ☒ Customizing Requests
- ☒ Workbench Requests
- ☐ Transport of Copies
- ☐ Relocations

Request Status

- ☒ Modifiable
- ☒ Released [Dropdown]

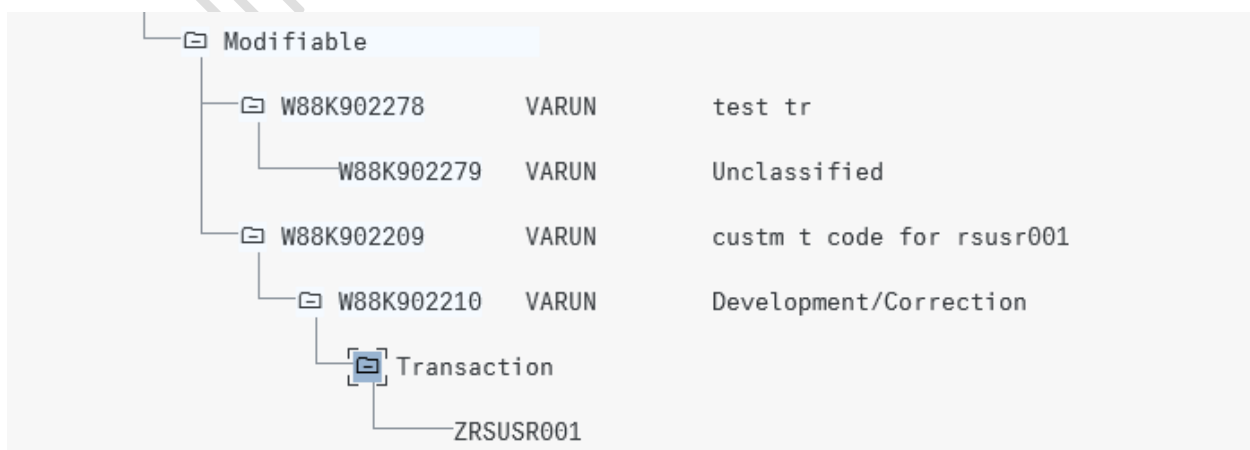
Global Information **Transport Proposals**

[Refresh Icon] [Status Bar: Red and Green Circles] 10 [Transports Button] [Repairs Button]

0

[Display Button]

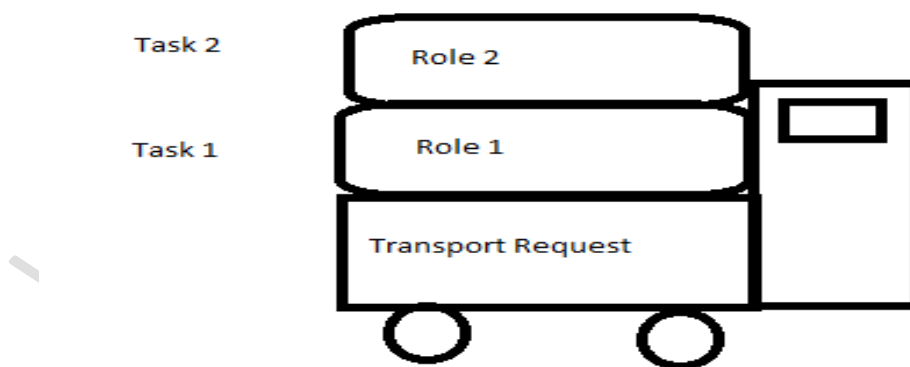
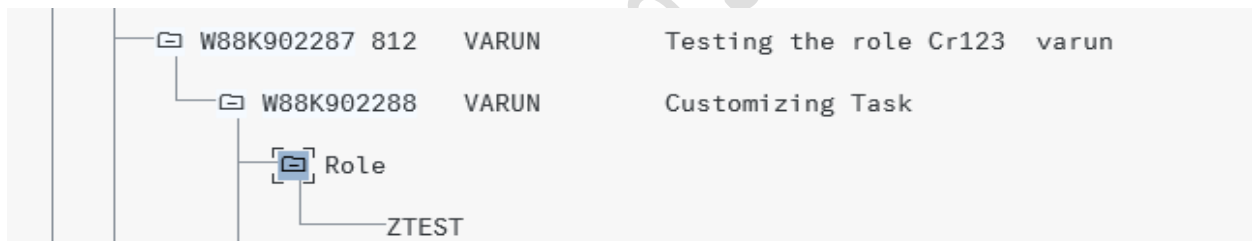
- ❖ **Workbench TR** => It carries **SAP standard** data. Ex – Abap code, T codes, programs



- ❖ **Customizing TR** => It carries **customized** data Ex – Business data, Users, Transactional data
- ❖ Transport request is a **group of tasks**.



- ❖ Task is space reserved for **one user**; on whose name the task is created only that user can load the data into **TR**.

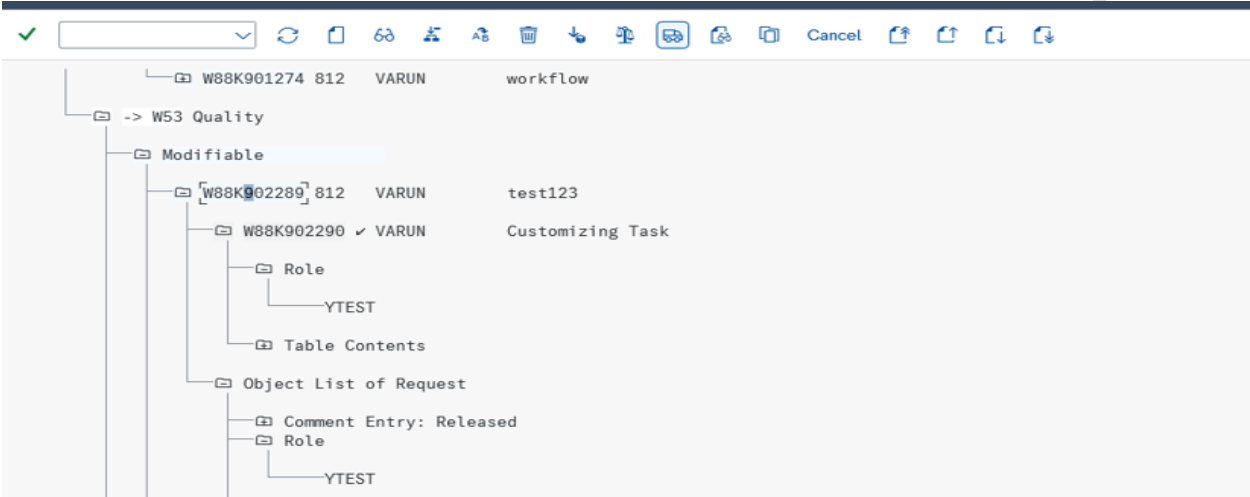


What are the **tables** related to **transport requests** in SAP?

- **E070**
- **E071**

Security Team duties regarding TMS.

- Creating a **Transport Request** (SE10 or SE09)
- Loading the **role** into **Transport Request** (PFCG)
- **Releasing** the Transport Request (SE10 or SE09)



- Intimate **Basis Team** that the TR has been **released** in dev system and request them to transport the TR into Quality.

Number	Request	RC	Owner	Short Text
27	W88K901025	▲	SHIRISHA	shirisha request number
28	W88K901027	▲	KARTHIK	Karthik's Request
29	W88K901029	▲	VAISHNAVI	This is Vaishnavi's request
30	W88K901035	▲	VARUN	varun Request Number
31	W88K901037	▲	AKASH	Akash request
32	W88K901039	▲	KAVYAV	kavya 2
33	W88K901047	▲	KARTHIK	Karthik's Currency Request
34	W88K901049	▲	LOHITYA	Lohitya's request
35	W88K901051	▲	SHIVALI	Shivali's Request
36	W88K901055	▲	KARTHIK	Karthik's Request (Country)
37	W88K901057	▲	SHAHEEN	Shaheen
38	W88K901067	▲	KAVYAV	kavya 3
39	W88K901073	▲	AKASH	Akash workbench
40	W88K901081	▲	ANKITA	Ankita Customizing request
41	W88K901142	▲	SUSHMITHA	SREEJA NEC MULTIPLICATION PROGRAM

Note => Before releasing the TR ensure **all the tasks** under that TR are **released**.

Only when **TR** is released from dev system, only then **basis team** can move that TR into **Quality system**. Else they cannot.



Role Modification

→ Single role

- ✧ Modifying menu by adding/removing T codes from Menu.
- ✧ Directly modifying authorization data.

→ Composite role

- ✧ Adding/removing single roles from Roles tab.

→ Derived role => Only Org Levels.

→ Master role

- ✧ This is modified similar to single role., Modifying menu by adding/removing T codes from Menu.
- ✧ Directly modifying authorization data.

Common Authorization Issues in SAP Security.

- Not able to login to SAP System.

- Wrong password, user locked, user expired, incorrect client, password expired—controlled by the parameter.

- Successfully logged into SAP System, but not able to execute the T code.

- Role is expired.
- Profile is not generated for the role.
- User comparison is not done.

Description			Menu			Workflow			Authorizations			User			MiniApps			Personalization		
Created			Last Changed			Last Profile Generation														
User			User			User														
Date			Date			Date														
Time	00:00:00		Time	00:00:00		Time	00:00:00													

Information About Authorization Profile	
Profile Name	
Profile Text	
Status	No authorization data exists

-Able to execute the T code, but not able to perform certain functions under that T code.

- User does not have access to appropriate authorization values required for the role assignment.

 You are not authorized to use transaction SU56	
--	--

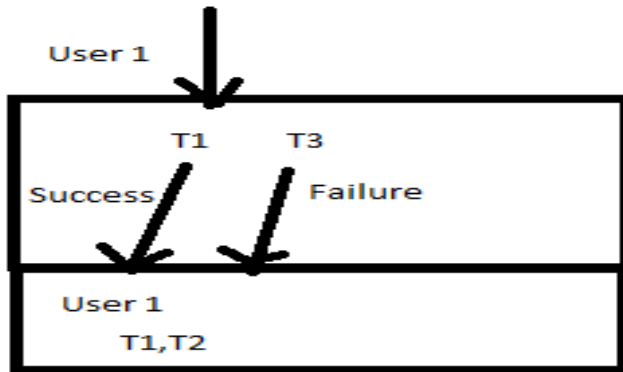
- Profile is not generated for the role.
- User comparison is not done.

T codes used in troubleshooting

SU53, SU56, ST01, STAUTHTRACE

1) SU53

- ❖ It evaluates the last authorization check, whether it is successful or failure.

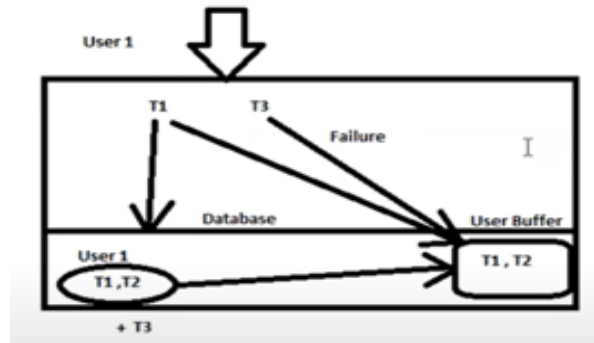


- ❖ When its failure even gives you the authorization values related to the failure.
- ❖ See the below picture it will which object is missing, T code, and their fields based on that we will give the authorization.

.....		
✓ ⚡	Authorization check failed	
✓	Date 30.07.2024 Time 11:08:37 Transaction	
✓	Authorization Obj. S_TCODE	Transaction Code Check at Transaction Start
	Authorization Field TCD	Transaction Code SM36
✓	Date 30.07.2024 Time 11:08:34 Transaction	SESSION_MANAGER
✓	Authorization Obj. P_RUNTIME	Personas Runtime Authorization object
	Authorization Field P_ACTVT_FW	Framework (UI Type) <Dummy>
	Authorization Field P_ACTVT_RT	Runtime Activity 01
	Authorization Field P_APP_ID	Application ID or Transaction Code <Dummy>

2)SU56

- ❖ It is a temporary memory created when user logs into SAP System and wipes out when the user logs out.
- ❖ When user executes any T code, System crosses check with the user buffer not with the DB.



❖ If there any problem related to the user buffer: -

- Ask user to logout and login to SAP System.
- Ask user to reset in user buffer.

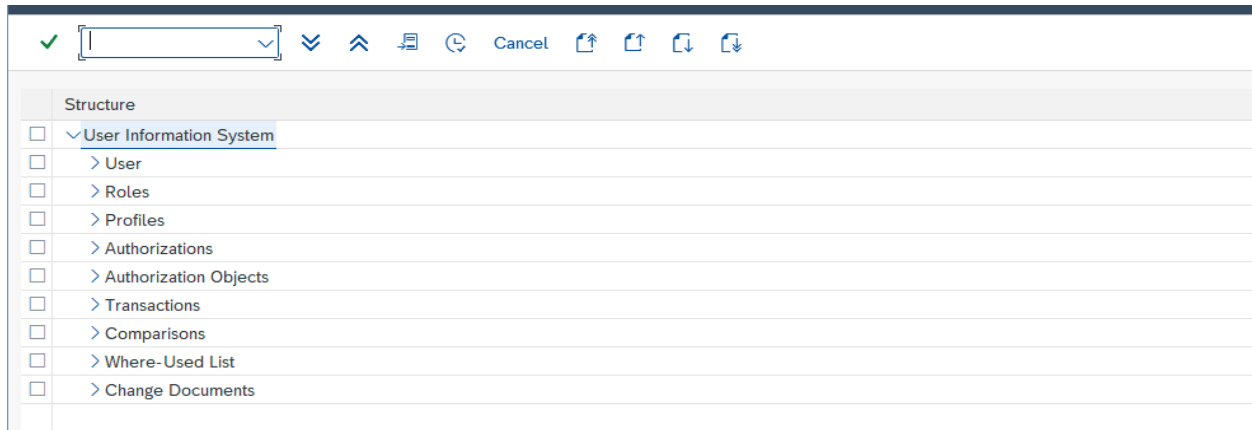
In the SU56 screen, there is an option called **Authorization Values**. Click on that and select the **User Buffer** option if any T-codes are not visible.

Number of Authorizations	4241

✓ User's Authorization Data VARUN	
✓ Object Class /AGG	AGRI: Authorization Objects
✓ Authorization Object /ACCGO/ACR	Application Rules - Condition Records
✓ Authorizat. &_SAP_ALL	Generated authorization for profile SAP_ALL
Profl. SAP_ALL	All SAP System authorizations
✓ Authorization Field /ACCGO/RID	Rule ID
*	
➤ Authorization Field ACTVT	Activity
➤ Authorization Object /ACCGO/AD	Application Document
➤ Authorization Object /ACCGO/APL	Application
➤ Authorization Object /ACCGO/APS	Application Archiving
➤ Authorization Object /ACCGO/ASG	Value Schedule Assignment to TSW Location
➤ Authorization Object /ACCGO/ASQ	Application Rules - Access Sequence Data

3)SUIM

It is a powerful tool for Security administrators to manage and audit user access and roles. It helps ensure that the right people have the right access and that security policies are being followed.



Important Options in SUIM:

1. User Information:

- ✓ **Users by Address Data:** Allows you to view user details, including email, and other contact information.
- ✓ **Users by Authorization Values:** Lets you find users based on specific authorization values.
- ✓ **Users by Role Assignment:** Provides a list of users assigned to specific roles, helping to track user permissions.



2. Role Information:

- ✓ **Roles by Complex Selection Criteria:** Enables searching for roles using various criteria like role name, description, or authorization objects.

- ✓ **Roles by User Assignment:** Lists all the roles assigned to specific users, which is useful for reviewing and managing user access.
- ✓ **Roles by Profile Assignment:** Shows roles associated with particular profiles.

 Roles by Complex Selection Criteria
 By Role Name
 By User Assignment
 By Transaction Assignment in Menu
 By Profile Assignment
 By Authorization Object
 By Authorization Values
 By Change Dates
 Search for Single Roles with Authorization Data

3. Authorization Information:

- ✓ **Profiles by Complex Selection Criteria:** Helps search for profiles based on different attributes, such as profile name or description.
- ✓ **Authorization Objects by Complex Selection Criteria:** Allows you to find authorization objects based on specific criteria.
- ✓ **Authorizations:** Lists all authorizations within the system, helping to manage and review authorization objects and their values.

 Profiles by Complex Selection Criteria
 By Profile Name or Text
 By Profiles Contained
 By Authorizations
 By Authorization Values
 By Last Change
 By Role
✓ Authorizations
 Authorizations by Complex Selection Criteria
 By Object

4. Change Documents:

- ✓ **Users:** Provides a history of changes made to user data, including who made the changes and what changes were made.
- ✓ **Roles:** Shows the change history for roles, helping track modifications over time.
- ✓ **Profiles:** Details the changes made to profiles, useful for auditing purposes.

☐	> User
☐	 For Role Assignment
☐	 For Roles
☐	 For Profiles
☐	 For Authorizations
☐	 For Authorization Defaults
☐	 For Security Policies
☐	 For CUA Settings

5. Other Tools:

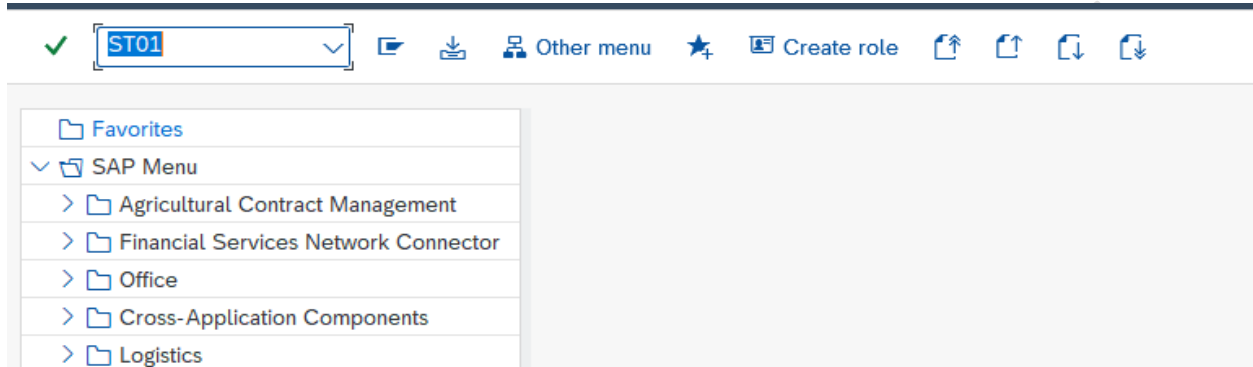
- ✓ **Where-Used List:** Shows where specific authorization objects or fields are used within the system.
- ✓ **Field Values:** Lists the possible values for a given authorization field.

☐	▼ Where-Used List
☐	> Roles
☐	> Profiles
☐	> Authorizations
☐	> Authorization Values
☐	> Authorization Objects
☐	> Security Policies

ST01 = System Trace

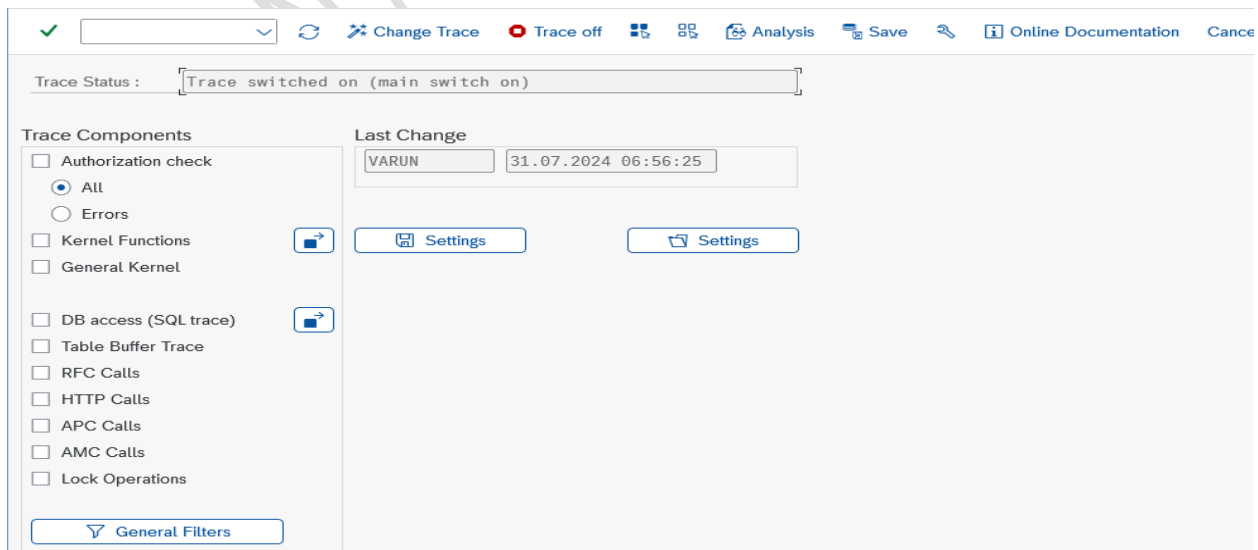
1) Access **the ST01** Transaction: -

- ❖ Go to the SAP Easy access **screen**.
- ❖ Enter the transaction code **ST01** in the command field and press **Enter**.



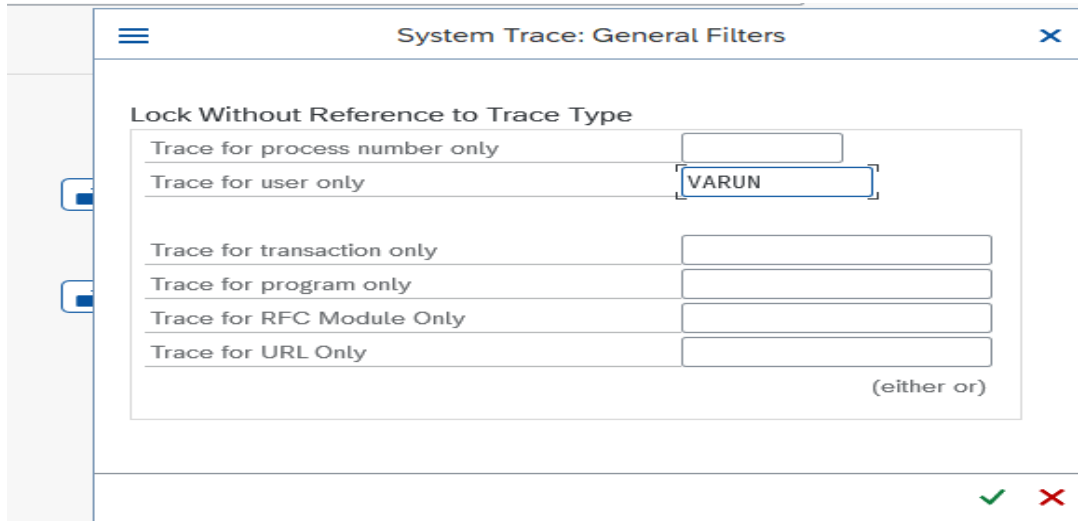
2) Select Trace Components: -

- In **the initial screen**, you will see options to select the **type of trace** you want to **activate**. Common options include:
 - ❖ **Authorization Check** => To monitor authorization checks.
 - ❖ **SQL Trace** => To log database queries.
 - ❖ **Buffer Trace** => To monitor buffer activities.
- Check the **boxes** for the components you want to **trace**.



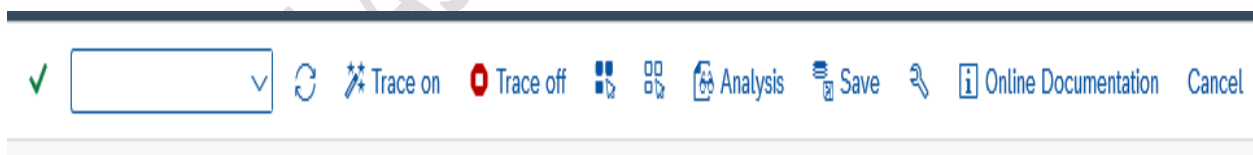
3) Set Trace Options (Optional): -

- You can **set filters** in the down what all are the **trace records**, such as:
 - ❖ **User IDs** => Specify particular users to trace.
 - ❖ **Transaction Codes** => Focus on specific transactions.
 - ❖ **Programs** => Monitor specific programs.



4) Activate the Trace: -

- ❖ Click on the **"Trace on"** button. The system will **start recording** activities based on your **selected** components and filters.



5) Perform the Activities to be Traced: -

- ❖ While the trace **is active**, login to that **User id** and perform the **actions** or **processes** you want to analyze.

6) Deactivate the Trace: -

- ❖ Once you have **captured** the necessary **information**, return to the ST01 screen, and click **"Trace off"** to stop recording.

7) Display and Analyze Trace Results: -

- ❖ Click on "**Analysis**" in ST01 **screen** and specify the user, date, and time to view the **recorded data**. Then execute to **see the Trace**.

General restrictions		Trace Records	
User name:	VARUN	☒ Authorization check	
Client:	812	☒ All	
Work Process:		☐ Errors	
Transaction:		☒ Kernel Functions	
Duration (>us):		☒ General Kernel	
Max. No. of Records	10.000	☒ DB access (SQL trace)	
From:	31.07.2024 / 07:52:19	☒ Table buffer trace	
To:	31.07.2024 / 08:02:19	☒ RFC calls	
Trans ID		☒ HTTP Calls	
Passport Full Context ID: connection ID:		☒ APC Calls	
		☒ AMC Calls	
		☒ Lock operations	

8) Troubleshoot and Analyze:

Client: 812 User: VARUN Transaction: SESSION_MANAGER Trans ID: 9E50DC0D35760000E00666FFC18163BE Start: 31.07.2024 07:31:14,125951 Finish: 31.07.2024 07:31:14,133065 No. of Records: 177 File Version: 3 EPP Full Context ID: 000C29A7FD041EDF93DF268F4C0B3186 EPP Connection ID: 00000000000000000000000000000000 EPP Call Counter: 0 Block Version: 54.200 First Block of Dialog Step Last Block in Dialog Step Work Process: 9 Process ID: 10.159				
hh:mm:ss:ms	Type	Lasts(us)	Object	Text
07:31:14,132	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: SAPLSMTR_NAVIGATION SESSION_A
Client: 812 User: VARUN Transaction: SESSION_MANAGER Trans ID: 9E50DC0D35760090E0066680631A99E2 Start: 31.07.2024 07:31:18,090265 Finish: 31.07.2024 07:31:18,122779 No. of Records: 1.882 File Version: 3 EPP Full Context ID: 000C29A7FD041EDF93DF268F4C0B3186 EPP Connection ID: 00000000000000000000000000000000 EPP Call Counter: 0 Block Version: 512.376 First Block of Dialog Step Last Block in Dialog Step Work Process: 9 Process ID: 10.159				
hh:mm:ss:ms	Type	Lasts(us)	Object	Text
07:31:18,100	AUTH		S_TCODE RC=0	tcode=SM36;TCD=SM36;type=TR;name= ;reason3=X;contextid=00007000000C29A7FD041EDF93DF
07:31:18,103	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: MENUSYST MEN
07:31:18,105	AUTH		S_BTCH_ADM RC=0	tcode=SM36;BTCADMIN=Y;type=TR;name=SM36;reason3=X;contextid=00007000000C29A7FD041ED
07:31:18,106	AUTH		S_BTCH_JOB RC=0	tcode=SM36;JOBGROUP=' ';JOBACTION=RELE;type=TR;name=SM36;reason3=X;contextid=000070
07:31:18,108	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: MENUSYST MEN
07:31:18,109	AUTH		S_BTCH_ADM RC=0	tcode=SM36;BTCADMIN=A;type=TR;name=SM36;reason3=X;contextid=00007000000C29A7FD041ED
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=01;type=TR;name=SM36;reason3=X;co
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=07;type=TR;name=SM36;reason3=X;co
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=04;type=TR;name=SM36;reason3=X;co

End of trace files reached.

- ❖ Use the **trace data** to identify **issues**, such as unauthorized access attempts, slow-performing SQL queries, or other system problems.

- ❖ Based on that the **security consultants** give the authorization, if fails by identify by using the **return codes in red color**.

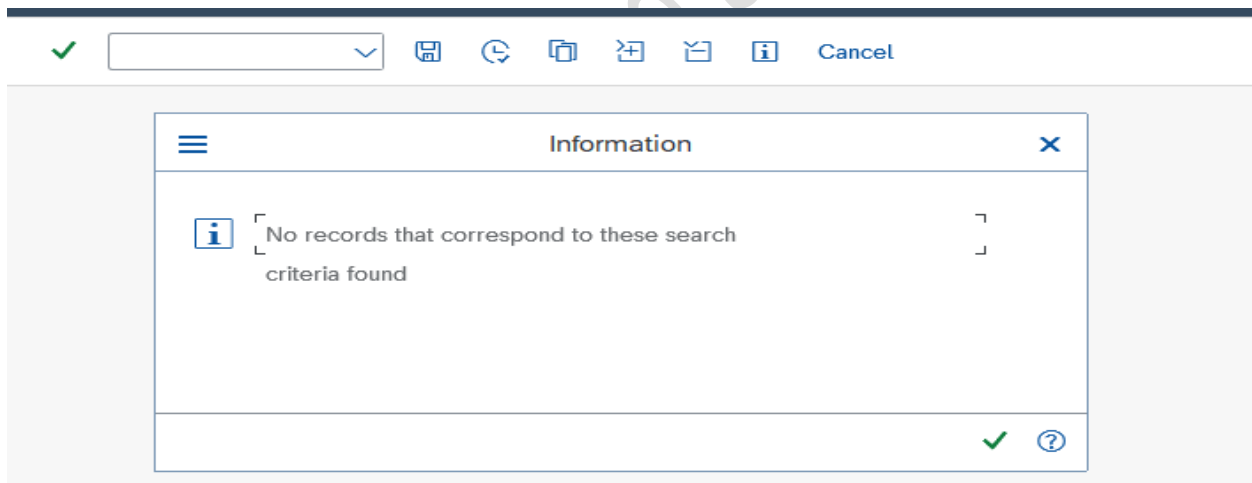
Note => It is important to **deactivate** the trace when not in use.

Return Codes

- **RC => 0** --- **Authorization** is successful.
- **RC => 4** --- User has required **authorization object**, but different **authorization Values**.
- **RC => 8** --- User does not have required **authorization values** in the user buffer.
- **RC => 12** --- User does not have **access to authorization object**.

If any 0 Records found

→ means if no trace display.



- ✓ **Incorrect** input values.
- ✓ User present in different **application server**.

Note => **ST01 trace** should be applied in the **local server only**. Hence before applying **ST01 trace**, ensure that **user** and **you** are in the same system.

AL08 => Users in each server.

SM51 => List of application servers.

STAUTHTRACE

- ⇒ Exclusive **authorization** Trace.
- ⇒ Trace is applied across all **application servers**.

Note => This T code is available in the updated version of SAP.

How to Use STAUTHTRACE:

✳ Access the Transaction:

- Enter **STAUTHTRACE** in the SAP command field and press Enter.

✳ Start the Trace:

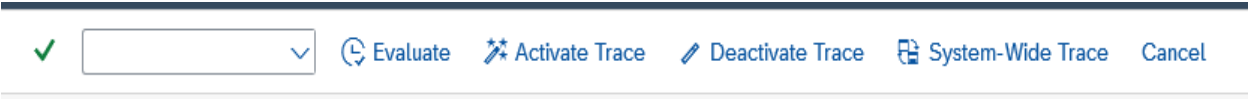
- Specify the **Username** and Click on "**Start Trace**" to begin recording **authorization checks**.

✳ Perform Activities:

- Inform the **user** to perform the **actions** that are causing **access issues**, such as trying to **execute** a transaction they cannot access.

✳ **Stop the Trace:**

- Once the **activities** are completed, go back to the **STAUTHTRACE** screen, and click **"Deactivate Trace"**.



✳ **View and analyze results:**

- Click on **"Display Trace"** to see the **results**, and review the **authorization**.

A screenshot of the STAUTHTRACE screen. At the top, there is a green checkmark icon, a dropdown menu, and several buttons: 'Evaluate', 'Activate Trace', 'Deactivate Trace', 'System-Wide Trace', and 'Cancel'. The 'Display Trace' button is highlighted.

Date	Date/Time	User	Type	Application Name	Program Name	Che...	Res...	Result	Addit...	Ac	En...	Object	Field 1	Valu...	Field 2	Valu...	Field 3	Valu...	Field...	Valu...	Field...
31....	10:49:02....	VAR1	Tran...	PFCG	SAPLSUSE		4	Authorization che...				S_TCODE	TCD	PFCG							
31....	10:49:02....	VAR1	Tran...	SESSION_MANA...	/PERSONAS/CL_A...		12	No authorization i...				P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...	01				
31....	10:49:02....	VAR1	Tran...	SESSION_MANA...	/PERSONAS/CL_A...		12	No authorization i...				P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...					
31....	10:49:05....	VAR1	Tran...	SM37	SAPLSUSE		4	Authorization che...				S_TCODE	TCD	SM37							
31....	10:49:08....	VAR1	Tran...	SM36	SAPLSUSE		4	Authorization che...				S_TCODE	TCD	SM36							
31....	10:49:09....	VAR1	Tran...	SU01	SAPLSUSE		0	Authorization che...				S_TCODE	TCD	SU01							
31....	10:49:09....	VAR1	Tran...	SU01	SAPLSUSE		0	Authorization che...				S_USER_GRP	CLASS	ACTVT							
31....	10:49:09....	VAR1	Tran...	SU01	/PERSONAS/CL_A...		12	No authorization i...				P_RUNTIME	P_APP_ID	SU01	P_AC...	01	P_AC...	01			
31....	10:49:09....	VAR1	Tran...	SU01	/PERSONAS/CL_A...		12	No authorization i...				P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...					

It displays the **details** about the **authorization objects**, **field values**, **program name** and the **return codes**.

Note => Click on **System Wide Trace** to view the trace for each application server. Apply the necessary filters, then click on **Execute**.

A screenshot of the STAUTHTRACE screen. At the top, there is a green checkmark icon, a dropdown menu, and several buttons: 'Evaluate', 'Activate Trace', 'Deactivate Trace', 'Local Trace', and 'Cancel'. The 'System Wide Trace' button is highlighted.

Trace Status

Logon Group

Server	Trace Status	User Name	Error...	Changed By	Date
☐ willsys88_W88_00	Undefined		☐		
☐ willsys88_W88_02	Undefined		☐		

Trace Options

Trace for user only

Trace for errors only ☐

Restrictions for the Evaluation

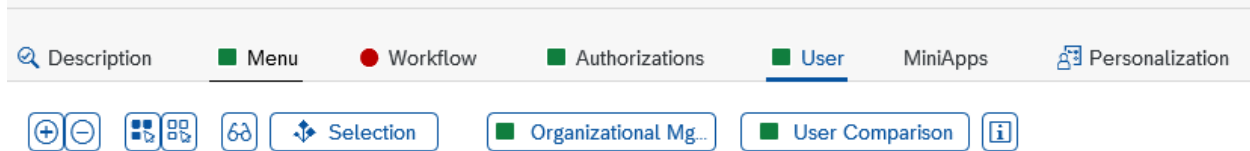
User to

From

To

PFUD**Mass User Comparison**

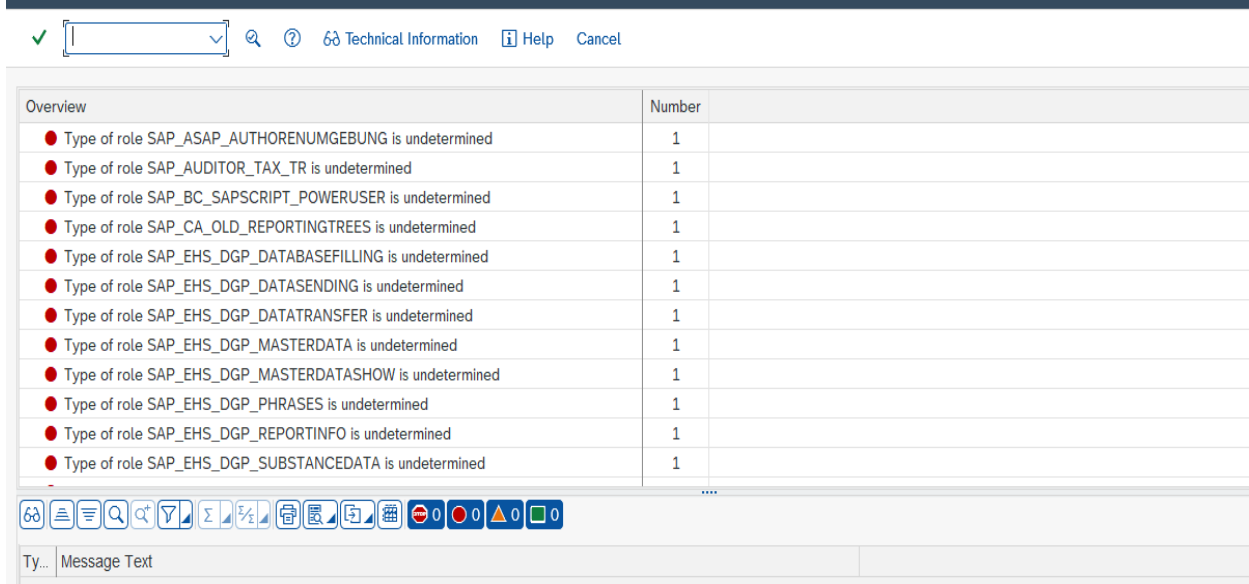
- ✓ For a **single role** we can do **user comparison** in PFCG -> User comparison Tab.



- ✓ If there is a requirement to perform **User comparison for 100 Roles** (Multiple roles), then we go to a T code called **PFUD**, to perform that.

- ✓ Click from **multiple role selection** and copy from **clipboard**.

✓ Then, click on **execute**.



The screenshot shows the SAP Security Manager interface. At the top, there is a search bar and buttons for 'Technical Information', 'Help', and 'Cancel'. Below this is a table with two columns: 'Overview' and 'Number'. The table lists 13 roles, all of which have a red dot icon in the 'Overview' column, indicating that their type is undetermined. The roles are:

Overview	Number
● Type of role SAP_ASAP_AUTHORENUMGEBUNG is undetermined	1
● Type of role SAP_AUDITOR_TAX_TR is undetermined	1
● Type of role SAP_BC_SAPSCRIPT_POWERUSER is undetermined	1
● Type of role SAP_CA_OLD_REPORTINGTREES is undetermined	1
● Type of role SAP_EHS_DGP_DATABASEFILLING is undetermined	1
● Type of role SAP_EHS_DGP_DATASENDING is undetermined	1
● Type of role SAP_EHS_DGP_DATATRANSFER is undetermined	1
● Type of role SAP_EHS_DGP_MASTERDATA is undetermined	1
● Type of role SAP_EHS_DGP_MASTERDATASHOW is undetermined	1
● Type of role SAP_EHS_DGP_PHRASES is undetermined	1
● Type of role SAP_EHS_DGP_REPORTINFO is undetermined	1
● Type of role SAP_EHS_DGP_SUBSTANCEDATA is undetermined	1

Below the table is a toolbar with various icons for navigation and actions. At the bottom, there is a 'Message Text' field.

What all important actions are performed in PFUD?

1) Profile matchup

- It ensures **profile** get assigned to user and the **role status is green** under **SU01**.

2) Composite role reconciliation

- Whenever **composite role is modified**, the role is **added or removed** from composite role, should reflect to **user as well**.

3) HR organizational management: reconciliation

- It ensures that **role** assigned to **position** or **org unit** of the user.

4) Cleanups

- It ensures that **expired roles** are removed from **SU01**, record of users.

Program related to **PFUD** => **PFCG_TIME_DEPENDENCY**

SM01 => Lock T code

EWZ5 => Mass User Lock

SU10 => **SU12**

IMPLEMENTATION PHASE

Every **Project** has two **Phases**.

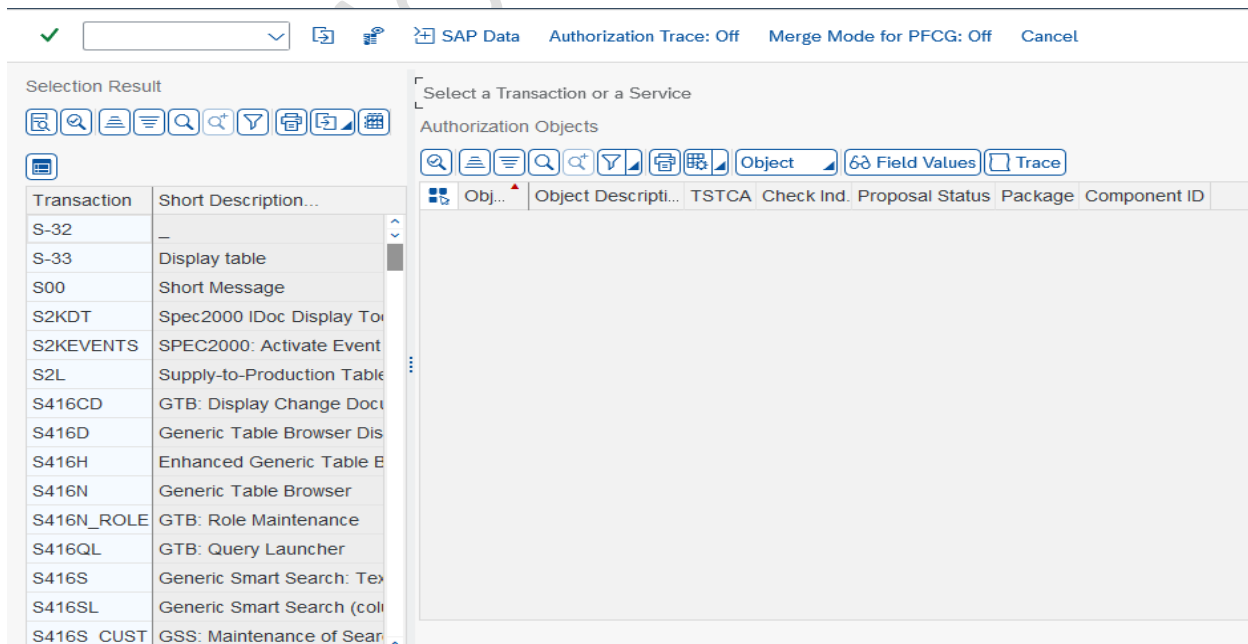
- ❖ Implementation => **Short** term.
- ❖ Support => **Long** term.

Types of projects

- ❖ Implementation => where there is **no SAP** at all decided to **implement SAP** in the company.
- ❖ Support => **already SAP** is there we look into **some issues**.
- ❖ Upgrade => **version** upgrade.
- ❖ Rollouts => extending **SAP** to other **business branches**.

What are the **roles** and **responsibilities** of Security Consultants in Implementation project.

- Roles => **PFCG** => **Authorization** data is pulled from SU24.
- The **image** below is a screenshot of the **SU24** transaction.

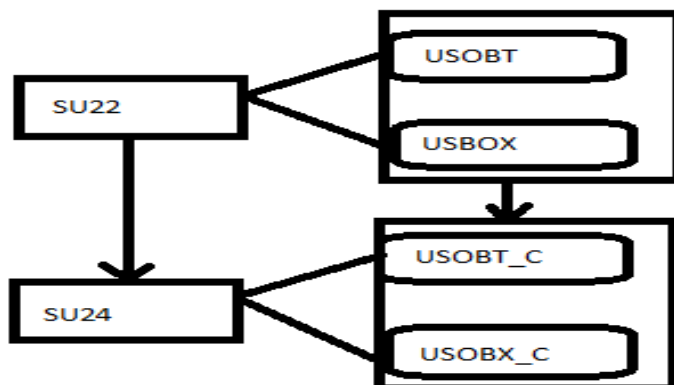


- Initially **if SU24 is empty**, then **PFCG T codes** does not work.
- We need to fill the **authorization data** into **SU24** by using the T code called **SU25**.
- The **image** below is a screenshot of the **SU25** transaction.

Actions to be Performed	Date	Time	User
> Installing the Profile Generator			
> Postprocess the Settings After Upgrading to a Higher Release			
> Transport Intf.			
> Adjust the Authorization Checks (Optional)			
> Manually Adjust Selected Roles			
> General Maintenance for Default Values			

SU25 => First Installation and Upgrade.

SU22 => SAP Standard Authorization Values related to a T code.



SU22 data is stored in Two Tables: -

1) **USOBT** => SAP Standard authorization Values related to a particular T code.

- ✓ SE16 => USOBT, the image below is a screenshot of the USOBT screen. If needed, you can apply the filters.

NAME	S*	to		→
TYPE		to		→
OBJECT		to		→
FIELD		to		→
LOW		to		→
HIGH		to		→
MODIFIER		to		→
MODDATE		to		→
MODTIME	00:00:00	to	00:00:00	→
Width of Output List	250			
Maximum No. of Hits	200			

- ✓ Then click on **execute** to see the columns like T code, auth object, field, and its range values.

SAP Data Browser: Table USOBT Select Entries 200

Table: USOBT
Displayed Fields: 9 of 9 Fixed Columns: 5 List Width: 0250

NAME	TYPE	OBJECT	FIELD	LOW	HIGH	MODIFIER	M
☐ /ACCGO/ABD_TYPE_SCEN	TR	S_TABU_DIS	ACTVT	02		SAP	0
☐ /ACCGO/ABD_TYPE_SCEN	TR	S_TABU_DIS	ACTVT	03		SAP	0
☐ /ACCGO/ABD_TYPE_SCEN	TR	S_TABU_DIS	DICBERCLS			SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	ACTVT	01		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	ACTVT	02		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	ACTVT	03		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	ACTVT	06		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	DEVCLASS			SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	OBJNAME			SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	OBJTYPE			SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_DEVELOP	P_GROUP			SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_SWITCH	ACTVT	02		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_SWITCH	ACTVT	03		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_SWITCH	ACTVT	07		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_SWITCH	OBJNAME	SFBF		SAP	0
☐ /ACCGO/ACTV_SWITCH	TR	S_SWITCH	OBJTYPE	DUMMY		SAP	0
☐ /ACCGO/AGR	TR	S_NUMBER	ACTVT	02		SAP	2

2) **USOBX** => Check **Indicators of Authorization object** related to a particular **T code** (OK Flag).

- ✓ The **image** below is a screenshot of the **USOBX** screen. If needed, you can apply the **filters** and also see the **difference** between both the **screens**.

NAME	S*	to		→
TYPE		to		→
OBJECT		to		→
MODIFIER		to		→
MODDATE		to		→
MODTIME	00:00:00	to	00:00:00	→
OKFLAG		to		→
Width of Output List	250			
Maximum No. of Hits	200			

- ✓ Then click on **execute** to see the columns like T code, auth object (cust) OKFLAG and its **Modifier name**.

Table: USOBX
Displayed Fields: 7 of 7 Fixed Columns: 3 List Width: 0250

	NAME	TYPE	OBJECT	MODIFIER	MODDATE	MODTIME	OKFLAG
☐	S-32	TR	S_TCODE	SAP	17.04.1998	16:07:00	X
☐	S-33	TR	S_TCODE	SAP	17.04.1998	16:07:00	X
☐	S00	TR	S_CTS_ADMI	SAP	12.01.2001	09:40:52	X
☐	S00	TR	S_DEVELOP	SAP	23.07.1996	17:07:55	X
☐	S00	TR	S_GUI	SAP	05.08.2003	11:22:02	X
☐	S00	TR	S_OC_DOC	SAP	09.07.1996	15:10:38	Y
☐	S00	TR	S_OC_FOLCR	SAP	09.07.1996	15:10:38	Y
☐	S00	TR	S_OC_ROLE	SAP	09.07.1996	15:10:39	Y
☐	S00	TR	S_OC_SEND	SAP	09.07.1996	15:10:40	Y
☐	S00	TR	S_OC_TCD	SAP	09.07.1996	15:10:41	Y
☐	S00	TR	S_RFC	SAP	18.10.2001	15:12:40	X
☐	S00	TR	S_TCODE	SAP	01.04.1996	12:17:16	X
☐	S001	TR	S_DEVELOP	SAP	18.08.1997	16:59:40	X
☐	S001	TR	S_SPO_DEV	SAP	03.11.1998	17:49:55	X

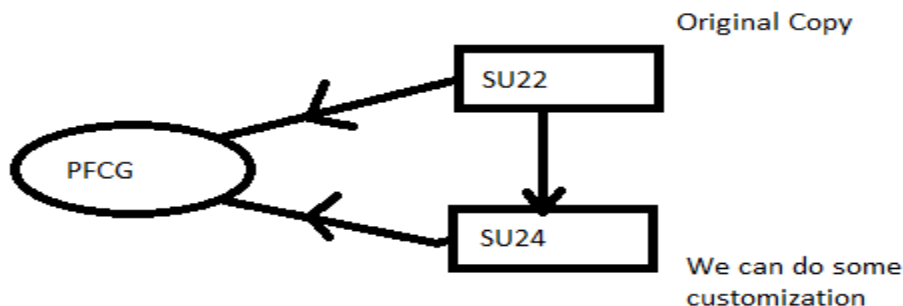
- ✓ Here **OKFLAG** contains the symbols called X, U, etc. and its description shown below.

Check flag	Short Descript.
N	No authorization check
X	Authorization check takes place
U	Not maintained
Y	Authorization check takes place; default values in USOBT
	Not maintained
V	Authorization check takes place, no default values

SU24 data is stored in Two Tables: -

- 1) **USOBT_C** => Customization + Standard.
- 2) **USOBX_C** => Standard + Customization

Note =>



- ✧ While the **role** is created, when **T codes** are added in the menu, **Authorization data** is pulled from SU24.
- ✧ For the **safety purpose** we have take a copy from **SU22 to SU24**, then when we added **T codes** in the menu, it will pull from **SU24** authorization data to **PFCG**.

Some Important Implementation T codes

SU20 => Authorization Field creation.

- The **image** below is a screenshot of the **SU20** Transaction.

The screenshot shows the SU20 Transaction screen. At the top, there is a toolbar with a green checkmark, a dropdown menu, a save icon, a refresh icon, a 'Sync. Field Buffer' button, an 'Authorization Field' button, an information icon, and a 'Cancel' button. Below the toolbar, the 'Standard Selection' section contains fields for 'Authorization Field Name', 'Data element', and 'Authorization Object', each with a selection icon. There is also a checkbox labeled 'Only Select OrgLevel Fields'. The 'Other Restrictions (Object Directory)' section contains fields for 'Package', 'Software Component', and 'Application Component', each with a selection icon.

SU21 => Authorization object creation.

- The **image** below is a screenshot of the **SU21** Transaction

The screenshot shows the SU21 Transaction screen. At the top, there is a toolbar with a green checkmark, a dropdown menu, a 'Regenerate SAP_ALL' button, and a 'Cancel' button. Below the toolbar, there is a row of icons for various functions. The main area is a table with three columns: 'Class/Object', 'Text', and 'Type'.

Class/Object	Text	Type
> _____	Objects with Invalid Object Class	Object C...
> /AGG	AGRI: Authorization Objects	Object C...
> /ARB	Object class for Ariba	Object C...
> /CPD	Commercial Project Management	Object C...
> /F15	/F15	Object C...
> /GEO	GEO.e Authorizations	Object C...
> /LSI	Financial accounting for leasing	Object C...
> /MTM	MtM Authorization Objects	Object C...

SU22 => Authorization data related to a T code.

SU24 => **Custom Authorization** data related to a T code.

SU25 => First Installation and **Upgrade**.

SM51 => List of Application **servers**.

AL08 => Users logged in to **all** servers.

SM04 => Users logged in to **local** servers.

SE16 => Table Display.

SM30 => Table Maintenance.

- The **image** below is a screenshot of the **SM30** Transaction.

✓ [Dropdown] Find Maintenance Dialog Cancel

Table/View [Text Field] [Search Icon]

Restrict Data Range

☒ No Restrictions

☐ Enter conditions

☐ Variant [Text Field]

[Display] [Maintain] [Transport] [Customizing]

SE38 => Abap editor, **create** and **edit** programs.

- The **image** below is a screenshot of the **SE38** Transaction.

✓ [Dropdown] [Icons] Debugging With Variant Variants Cancel

Program [Text Field] [Search Icon] [Create]

Subobjects

☒ Source Code

☐ Variants

☐ Attributes

☐ Documentation

☐ Text elements

[Display] [Change]

Some Important Authorization objects

1) User administration

- ❖ S_USER_GRP
- ❖ S_USER_AGR
- ❖ S_USER_PRO
- ❖ S_USER_AUTH
- ❖ S_USER_SAS
- ❖ S_USER_SYS

2) Role administration

- * S_USER_AGR
- * S_USER_TCD
- * S_USER_PRO
- * S_USER_VAL

3) Abap

- ❖ S_DEVELOP
- ❖ S_PROGRAM

4) Archiving

- * S_ARCHIVE

5) T Code

- * S_TCODE

6) Table Access

- ❖ S_TABU_DIS
- ❖ S_TABU_NAM
- ❖ S_TABU_LIN
- ❖ S_TABU_CLI

7) Spool Admin

- * S_SPO_DEV
- * S_SPO_PAGE
- * S_SPO_ACT

8) TMS

- * S_TRANSPRT
- * S_CTS_ADMI
- * S_CTS_LANG
- * S_CTS_TADM

10) Background Admin

- ❖ S_BTCH_ADM
- ❖ S_BTCH_JOB
- ❖ S_BTCH_NAM
- ❖ S_BTCH_MONI

11) File Access

- * S_DATASET
- * S_PATH

Note => S_CTS_SADM: CTS system administration (used **instead** of S_CTS_ADMI in **some** cases)

→ In **SU20** famous field is **TCD/CLASS**.

→ **Never delete** standard authorization or **modified authorization field**.

3.200 authorization fields selected



Field Name	Short Description
TCD	Transaction Code
TCODE	Transaction Code

→ **OSS** => online service system, it will come while **creating** authorization object.

SU25 => First Installation and upgrade.

- It helps to **adjust** and **update** authorization data, specifically the authorization **default values**, to ensure they are compatible with the updated **version** or **system**.

What are all **the actions** performed in **SU25** means: -



Actions to be Performed	Date	Time	User
✓ Installing the Profile Generator			
↳ Initial fill of customer tables was performed (1)	16.04.2012	16:18:24	HUETT
✓ Postprocess the Settings After Upgrading to a Higher Release			
↳ Automatic Comparison with SU22 Data (2a)	18.07.2024	17:37:36	ANKITA00
↳ Modification Comparison with SU22 Data (2b)	18.07.2024	17:38:46	ANKITA00
↳ Search for Obsolete Applications (2d)	21.07.2024	20:36:13	ANKITA00
↳ Update of Application Groups in Role Menu			
↳ Roles to Be Checked (2c)	16.07.2024	12:49:19	VAISHNAVI
✓ Transport Intf.			
↳ Transport the Customer Tables (3)	16.07.2024	11:49:05	VAISHNAVI
✓ Adjust the Authorization Checks (Optional)			
↳ Check Indicators in Applications (SU24)			
↳ Deactivate Authorization Object Globally			
↳ Transaction Start Authorization Check (SE97)			
↳ Comparison of Switchable Authorization Checks (SACF)			
↳ Comparison of Generic Whitelists (SLDW)			

✳ When we click on **initial fill of customer tables**, **SU22** data will copy to **SU24**, All the values are overwritten from SU22 to SU24, by losing some customization. = **Step 1**

- ✓ If we do not lose **the customization**, whatever the data is there in **SU22** same will be **copied**, how means by doing remaining steps, 2A, 2B, 2D, 2C.
= **Step 2**

✓	Postprocess the Settings After Upgrading to a Higher Release
⌚	Automatic Comparison with SU22 Data (2a)
⌚	Modification Comparison with SU22 Data (2b)
⌚	Search for Obsolete Applications (2d)
⌚	Update of Application Groups in Role Menu
⌚	Roles to Be Checked (2c)

- **2A (Compare with SAP Values)** => It will **compare** the SAP values between **SU22** and **SU24**. It will check with the **standard** application not customized one.
- ➔ **2B (Compare affected transactions)** => While we are **running** this manual comparison is required it will ask u like which **Authorizations** you want to proceed whether u want an **SAP** suggested or customized, most of the cases we proceed with customized one that means **SU24** data only.
- ✧ **2D (Displayed Changed T codes)** => When we **click** on this, it displays the role name, description, old T-code (to be deleted), and newly **introduced T-code**. For example, well-established **roles** from past years might have authorization fields and objects tied to the old T-code. Removing these without understanding could affect other T-codes that depend on the **same objects**.
- **2C (Roles to be checked)** => It provides the **lists of roles** that need to be **reviewed** for changes in **authorizations**, due to updates in transaction codes. It helps **identify roles** that may require adjustments to maintain proper access **control**.
- ✓ If anything made changes in **SU24**, then click on 3) **Transport the customer tables**. = **Step 3**

- ✓ After this, **Check Indicators** in SU24 = **Step 4**
- ✓ **Deactivate Authorization objects globally**, In SU21 there is a folder called **AAA** – obsolete authorization objects. = **Step 5**
- ✓ Copy data from **old** profiles. = **Step 6**

Note => If we are performing **first step** called Initial fill the Customer tables, then remaining **steps are not needed**.

- ✂ **SA38** => T code is used to execute the programs.
- ✂ **SUPC** => Mass generation of profiles for Multiple roles.
- ✂ Organizational fields are created by using the report called **PFCG_ORGFIELD_CREATE**.
- ✂ **TCDCOUPLES** => List of called Transactions.
- ✂ **T000** => Table related to the client.
- ✂ **TSTC** => T code for Table checking.
- ✂ **TSTCT** => Text for the T codes.
- ✧ **SUKRI** => It is used to maintain the list of Transactions.
- ✧ **USKRIA** => It is used to maintain the list of Authorizations.
- ✧ **TBRG** => Authorization group programs stored in this table.
- ✧ **TDDAT** => Table related authorization group.
- ✧ **TRDIR** => Program for authorization group relation.
- ✧ **DBTABLOG** => Log records of table changes.
- ✧ **RFCDES** => Rfc destinations.
- ✧ **TADIR** => Stores repository objects.
- ✧ **TOBJ** => All the authorization objects exist in the System.
- ✧ **TOBC** => Details of authorization objects class.
- ✧ **TOBCT** => Text for authorization object class.
- ✧ Chang documents are stored in **USH***

THANK YOU

